Optimization Methods

Seminar 03

Gradient Descent and Automatic Differentiation

Theory handout

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These notes collect the theory of Seminar 3: the definitions for the colloquium, Jensen's inequality, gradient descent with a constant step and with backtracking, convergence rates and reverse-mode automatic differentiation. Every statement comes with its hypotheses and, where the colloquium may ask for it, a complete proof, and the seminar problems appear with full solutions.

Notation.

Vectors are columns, $x^\top y$ is the inner product and $\|\cdot\|$ the Euclidean norm unless a subscript says otherwise. The function f is defined on $\mathbb{R}^n$, $f^* = \inf_x f(x)$ is the optimal value and x^* a minimizer when one exists. The iterates are $x_0, x_1, \dots$, the step is α or α_k, the direction d or d_k, and x^+ is the next iterate inside a proof. The constants of strong convexity and smoothness are μ and L, and $\kappa = L/\mu$. For symmetric matrices $A \preceq B$ means that $B - A$ is positive semidefinite. In automatic differentiation $\dot{v}$ is a tangent and $\bar{v} = \partial z/\partial v$ an adjoint.

1 Definitions

At the colloquium definitions come first, and a definition counts only with its hypotheses. Each block below is the form to reproduce; the short paragraphs between them are consequences, not part of the definitions.

Definition: Local and global minimum. Let $f : \mathbb{R}^n \to \mathbb{R}$. A point x^* is a **global minimizer** of f if $f(x^*) \le f(x)$ for all $x \in \mathbb{R}^n$. It is a **local minimizer** if there is $\varepsilon > 0$ such that $f(x^*) \le f(x)$ for all x with $\|x - x^*\| < \varepsilon$. A minimizer is **strict** if the inequality is strict for every $x \neq x^*$ in the corresponding set. The value $f(x^*)$ is then a global or local **minimum**, and $f^* = \inf_x f(x)$ is the optimal value.

The optimal value may be $-\infty$, and it may be finite without being attained: $f(x) = e^x$ on $\mathbb{R}$ has $f^* = 0$ and no minimizer.

Definition: Stationary point. Let $f : \mathbb{R}^n \to \mathbb{R}$ be differentiable at x. The point x is **stationary** if $\nabla f(x) = 0$.

A local minimizer of a differentiable function is stationary: if $\nabla f(x^*) \neq 0$, then $d = -\nabla f(x^*)$ is a descent direction (defined below) and f decreases along it. The converse fails, since a stationary point may be a local maximizer or a saddle point. For a convex differentiable f every stationary point is a global minimizer: put $\nabla f(x^*) = 0$ into the tangent inequality of the next section.

Definition: Convex set. A set $C \subseteq \mathbb{R}^n$ is **convex** if $(1-t)x + ty \in C$ for all $x, y \in C$ and all $t \in [0,1]$, that is, if it contains the segment between any two of its points.

Definition: Convex and strictly convex function. Let $C \subseteq \mathbb{R}^n$ be a convex set, usually $C = \mathbb{R}^n$. A function $f : C \to \mathbb{R}$ is **convex** if it satisfies Jensen's inequality

$$f\big((1-t)x + ty\big) \le (1-t)f(x) + tf(y) \quad \text{for all } x, y \in C,\ t \in [0,1].$$

It is **strictly convex** if the inequality is strict whenever $x \neq y$ and $t \in (0,1)$. It is **concave** if $-f$ is convex.

Definition: μ-strong convexity. Let $\mu > 0$. A function $f : \mathbb{R}^n \to \mathbb{R}$ is **μ-strongly convex**

if $f(x) - \frac{\mu}{2}\|x\|^2$ is convex. Equivalently, Jensen's inequality holds with a quadratic margin:

$$f\big((1-t)x + ty\big) \le (1-t)f(x) + tf(y) - \frac{\mu}{2}t(1-t)\|x-y\|^2 \quad \text{for all } x, y,\ t \in [0,1].$$

If f is differentiable, this is equivalent to the first-order form

$$f(y) \ge f(x) + \nabla f(x)^\top (y-x) + \frac{\mu}{2}\|y-x\|^2 \quad \text{for all } x, y,$$

and if f is twice differentiable, to $\nabla^2 f(x) \succeq \mu I$ for all x.

The forms agree because $(1-t)\|x\|^2 + t\|y\|^2 - \|(1-t)x + ty\|^2 = t(1-t)\|x-y\|^2$ (the identity of the next section), and the first-order form is the tangent inequality for $f - \frac{\mu}{2}\|\cdot\|^2$. A strongly convex function is strictly convex and has exactly one minimizer. Strict convexity is weaker: $f(x) = \log(1+e^x)$ has $f''(x) = \sigma(x)\big(1-\sigma(x)\big) > 0$ with $\sigma(x) = 1/(1+e^{-x})$, but $f''(x) \to 0$ as $|x| \to \infty$, so no $\mu > 0$ works, and this f has no minimizer at all.

Definition: L-smoothness. Let $L > 0$. A differentiable function $f : \mathbb{R}^n \to \mathbb{R}$ is **L-smooth** if its gradient is L-Lipschitz:

$$\|\nabla f(x) - \nabla f(y)\| \le L\|x-y\| \quad \text{for all } x, y.$$

If f is twice continuously differentiable, this is equivalent to $-LI \preceq \nabla^2 f(x) \preceq LI$ for all x. For a convex f the left inequality is automatic, and the condition reads $\nabla^2 f(x) \preceq LI$.

The lecture uses the two names, L-smooth and L-Lipschitz gradient, for the same property. The property that the convergence proofs use is the descent lemma of the gradient descent section: $f(y) \le f(x) + \nabla f(x)^\top (y-x) + \frac{L}{2}\|y-x\|^2$.

Definition: Condition number. Let f be μ-strongly convex and L-smooth, with μ the largest and L the smallest constants for which this holds. The **condition number** of f is $\kappa = L/\mu$, and $\kappa \ge 1$. For a quadratic $f(x) = \frac{1}{2}x^\top Q x - b^\top x$ with symmetric $Q \succ 0$ the best constants are $\mu = \lambda_{\min}(Q)$ and $L = \lambda_{\max}(Q)$, so $\kappa = \lambda_{\max}(Q)/\lambda_{\min}(Q)$.

The level sets of such a quadratic are ellipsoids, and their longest and shortest semi-axes are in the ratio $\sqrt{\kappa}$. A large κ means a long narrow valley.

Definition: Descent direction. Let $f : \mathbb{R}^n \to \mathbb{R}$ be differentiable at x. A vector $d \in \mathbb{R}^n$ is a **descent direction** at x if $\nabla f(x)^\top d < 0$. Then $f(x + \alpha d) < f(x)$ for all sufficiently small $\alpha > 0$, since $f(x + \alpha d) = f(x) + \alpha\, \nabla f(x)^\top d + o(\alpha)$.

If $\nabla f(x) \neq 0$, the vector $d = -\nabla f(x)$ is a descent direction: $\nabla f(x)^\top d = -\|\nabla f(x)\|^2 < 0$. The condition is sufficient for local decrease, not necessary: at the maximizer 0 of $f(x) = -x^2$ every $d \neq 0$ decreases f, while $f'(0)\, d = 0$.

Definition: Polyak–Łojasiewicz condition. Let $f : \mathbb{R}^n \to \mathbb{R}$ be differentiable with $f^* = \inf_x f(x) > -\infty$, and let $\mu > 0$. The function f satisfies the **Polyak–Łojasiewicz (PL) condition** with constant μ if

$$\frac{1}{2}\|\nabla f(x)\|^2 \ge \mu\big(f(x) - f^*\big) \quad \text{for all } x.$$

Under PL every stationary point is a global minimizer, since $\nabla f(x) = 0$ forces $f(x) = f^*$. A

μ-strongly convex differentiable function satisfies PL with the same μ: minimizing the right-hand side of the first-order form over y, at $y = x - \nabla f(x)/\mu$, gives $f^* \geq f(x) - \frac{1}{2\mu}\|\nabla f(x)\|^2$. The converse fails, and PL does not even need convexity: $f(x) = x^2 + 3\sin^2 x$ satisfies PL with some $\mu > 0$, while $f''(\pi/2) = 2 + 6\cos\pi < 0$.

2 Jensen's inequality

Jensen's inequality is the definition of convexity. The usual hypotheses of the convergence results are the same inequality with a quadratic margin: a lower one for strong convexity, an upper one for smoothness.

2.1 Reading the inequality

Convexity of f on a convex set is Jensen's inequality

$$f\big((1-t)x + ty\big) \leq (1-t)f(x) + tf(y), \qquad t \in [0,1].$$

As t runs from 0 to 1, the point $(1-t)x + ty$ runs over the segment from x to y, so the left-hand side is the graph of f over the segment. The right-hand side applies the same weights to the values: it is the chord from $(x, f(x))$ to $(y, f(y))$. Convexity says exactly that the chord never dips below the graph. Differentiability plays no role here: $f(x) = |x|$ is convex by the triangle inequality $|(1-t)x + ty| \leq (1-t)|x| + t|y|$, yet it is not differentiable at 0.

The restriction $t \in [0,1]$ matters. For other t the point $(1-t)x + ty$ still lies on the line through x and y, but not between them, and the inequality fails even for $f(x) = x^2$. With $x = 0$, $y = 1$ and $t = 2$ the point is 2, the left-hand side is $f(2) = 4$, and the right-hand side is $(1-t)f(x) + tf(y) = 2$.

> **Proposition: The identity for x^2.** For all real x, y and t,
>
> $$(1-t)x^2 + ty^2 - \big((1-t)x + ty\big)^2 = t(1-t)(x-y)^2.$$

> *Proof.* Expand $\big((1-t)x+ty\big)^2 = (1-t)^2x^2 + 2t(1-t)xy + t^2y^2$ and use $(1-t) - (1-t)^2 = t(1-t)$ and $t - t^2 = t(1-t)$: the difference is $t(1-t)(x^2 - 2xy + y^2) = t(1-t)(x-y)^2$.

For $t \in [0,1]$ the right-hand side is nonnegative, so x^2 is convex, and it vanishes only for $t \in \{0,1\}$ or $x = y$, so x^2 is strictly convex. For $t \notin [0,1]$ the factor $t(1-t)$ is negative and the inequality fails for every $x \neq y$; in the example above $t(1-t)(x-y)^2 = 2 - 4 < 0$. The same computation with inner products gives, for vectors,

$$(1-t)\|x\|^2 + t\|y\|^2 - \|(1-t)x + ty\|^2 = t(1-t)\|x-y\|^2,$$

and for $f(x) = \frac{1}{2}x^\top Qx$ with symmetric Q the gap between chord and graph is $\frac{1}{2}t(1-t)(x-y)^\top Q(x-y)$.

2.2 Three equivalent characterizations

The chord is the definition. Two other tests are equivalent to it, and each needs smoothness that the definition does not.

Characterization	Needs	Meaning
Chord (Jensen): $f\big((1-t)x+ty\big) \le (1-t)f(x)+tf(y)$	a convex domain, no derivatives	the graph lies on or below every chord
Tangent: $f(y) \ge f(x)+\nabla f(x)^\top (y-x)$	f differentiable on an open convex set	the graph lies on or above every tangent plane; the linear model is a global lower bound
Curvature: $\nabla^2 f(x) \succeq 0$	f twice differentiable on an open convex set	$d^\top \nabla^2 f(x)\, d \ge 0$ for every d: the graph curves upward along every line

Theorem: Characterizations of convexity. Let $C \subseteq \mathbb{R}^n$ be open and convex. If f is differentiable on C, it is convex if and only if $f(y) \ge f(x)+\nabla f(x)^\top (y-x)$ for all $x, y \in C$. If f is twice differentiable on C, it is convex if and only if $\nabla^2 f(x) \succeq 0$ for all $x \in C$.

Proof. From the chord to the tangent: for $t \in (0,1]$ the chord inequality gives $f\big(x+t(y-x)\big)-f(x) \le t\big(f(y)-f(x)\big)$; divide by t and let $t \to 0$ to get $\nabla f(x)^\top (y-x) \le f(y)-f(x)$. From the tangent to the chord: for $z=(1-t)x+ty$ the tangent inequality at z gives $f(x) \ge f(z)+\nabla f(z)^\top (x-z)$ and $f(y) \ge f(z)+\nabla f(z)^\top (y-z)$; multiply by $1-t$ and t and add, and the gradient terms cancel because $(1-t)(x-z)+t(y-z)=0$. Curvature: f is convex if and only if every restriction $\varphi(s)=f(x+sd)$ to a line is convex, a twice differentiable function of one variable is convex if and only if $\varphi'' \ge 0$, and $\varphi''(s)=d^\top \nabla^2 f(x+sd)\, d$.

For strict convexity only one direction survives: $\nabla^2 f(x) \succ 0$ for all x implies strict convexity, but $f(x)=|x|^3$ is strictly convex while $f''(x)=6|x|$ vanishes at 0.

2.3 Finite and expectation forms

Theorem: Jensen's inequality for several points. Let f be convex on a convex set C, let $x_1, \dots, x_N \in C$, and let $t_1, \dots, t_N \ge 0$ with $\sum_i t_i = 1$. Then

$$f\Big(\sum_{i=1}^{N} t_i x_i\Big) \le \sum_{i=1}^{N} t_i f(x_i).$$

Proof. Induction on N. For $N=1$ both sides are $f(x_1)$, and $N=2$ is the definition. Let $N \ge 3$ and assume the claim for $N-1$ points. If $t_N = 1$, all other weights vanish and there is nothing to prove. Otherwise

$$\sum_{i=1}^{N} t_i x_i = (1-t_N)\, y + t_N x_N, \qquad y = \sum_{i=1}^{N-1} \frac{t_i}{1-t_N}\, x_i,$$

and y is a convex combination of $N-1$ points of C: its weights are nonnegative and sum to 1, so $y \in C$. The definition and then the induction hypothesis give

$$f\Big(\sum_{i=1}^{N} t_i x_i\Big) \le (1-t_N) f(y) + t_N f(x_N) \le \sum_{i=1}^{N-1} t_i f(x_i) + t_N f(x_N).$$

Theorem: Jensen's inequality for expectations. Let $f : \mathbb{R}^n \to \mathbb{R}$ be convex and let X

be a random vector in $\mathbb{R}^n$ with $\mathbb{E}\|X\| < \infty$ and $\mathbb{E}|f(X)| < \infty$. Then

$$f(\mathbb{E}X) \le \mathbb{E}f(X).$$

Proof. Put $m = \mathbb{E}X$. If f is differentiable, the tangent inequality at m reads $f(X) \ge f(m) + \nabla f(m)^\top (X - m)$; take expectations and use $\mathbb{E}(X - m) = 0$. Without differentiability the same argument works with a subgradient at m, a vector g with $f(y) \ge f(m) + g^\top (y - m)$ for all y, which exists at every point for a convex $f : \mathbb{R}^n \to \mathbb{R}$. A random vector that takes the value x_i with probability t_i gives back the finite form.

2.4 Concave functions and the AM–GM inequality

Trap. For a concave g every inequality above reverses: $g\big(\sum_i t_i x_i\big) \ge \sum_i t_i g(x_i)$ and $g(\mathbb{E}X) \ge \mathbb{E}g(X)$. The logarithm is concave, so $\mathbb{E}\log X \le \log \mathbb{E}X$ for a positive random variable X, and not the other way round.

Proposition: AM–GM for two numbers. For $a, b > 0$,

$$\sqrt{ab} \le \frac{a+b}{2},$$

with equality if and only if $a = b$.

Proof. The function $-\log$ is strictly convex on the convex set $(0, \infty)$, since $(-\log x)'' = 1/x^2 > 0$. The chord inequality at $t = \frac{1}{2}$ gives

$$-\log \frac{a+b}{2} \le \frac{1}{2}\big(-\log a - \log b\big) = -\log \sqrt{ab},$$

strictly unless $a = b$. Multiply by -1 and apply the increasing function exp.

Example. $a = 1$, $b = 9$: $\sqrt{ab} = 3 \le 5 = \frac{a+b}{2}$. With the finite form the same argument gives the weighted version $\prod_i a_i^{t_i} \le \sum_i t_i a_i$ for $a_i > 0$, $t_i \ge 0$, $\sum_i t_i = 1$.

2.5 A counterexample

Example. Take $f(x) = x^3$ on $\mathbb{R}$ with $x = -2$, $y = 1$ and $t = 0.5$. The point is $(1-t)x + ty = -0.5$, the left-hand side is $f(-0.5) = -0.125$, and the right-hand side is $(1-t)x^3 + ty^3 = -3.5$. Since $-0.125 > -3.5$, the inequality fails, and x^3 is not convex on $\mathbb{R}$.

One pair and one t disprove convexity; no number of successful checks proves it. On $[0, \infty)$ the function x^3 is convex, since $f''(x) = 6x \ge 0$ there, so a random check that draws 10000 pairs from $[0, 2]$ finds 0 violations and says nothing about negative arguments. Convexity is proved with the definition or with one of the characterizations above.

2.6 Quadratic margins on both sides

In the language of the chord, μ and L bound the gap between the chord and the graph from below and from above. If f is μ-strongly convex and L-smooth, then for all x, y and $t \in [0,1]$

$$\frac{\mu}{2}\,t(1-t)\|x-y\|^2 \le (1-t)f(x) + tf(y) - f\big((1-t)x+ty\big) \le \frac{L}{2}\,t(1-t)\|x-y\|^2,$$

and in first-order form

$$\frac{\mu}{2}\|y-x\|^2 \le f(y) - f(x) - \nabla f(x)^\top (y-x) \le \frac{L}{2}\|y-x\|^2.$$

The left inequalities are the definition of strong convexity. The right one in first-order form is the descent lemma, and its chord version follows by writing the descent lemma at $z = (1-t)x + ty$ for the points x and y and adding with weights $1-t$ and t. For a convex differentiable f each right inequality is in fact equivalent to L-smoothness.

For $f(x) = \frac{1}{2}x^\top Q x$ the gap between chord and graph equals $\frac{1}{2}\,t(1-t)(x-y)^\top Q(x-y)$. It lies between the two bounds with $\mu = \lambda_{\min}(Q)$ and $L = \lambda_{\max}(Q)$ and touches each of them when $x-y$ is a corresponding eigenvector. The ratio $\kappa = L/\mu$ says how far apart the two parabolas are, and the next sections show that it controls the number of gradient descent iterations.

> **Check yourself.** Apply Jensen's inequality to $f(x) = x^2$ and a random variable X with finite variance. What does it say, and what is the gap between the two sides?
>
> **Answer.** It says $(\mathbb{E}X)^2 \le \mathbb{E}X^2$. The gap is $\mathbb{E}X^2 - (\mathbb{E}X)^2 = \operatorname{Var} X \ge 0$, and it vanishes only when X is constant with probability one. If $X = x$ with probability $1-t$ and $X = y$ with probability t, then $\operatorname{Var} X = t(1-t)(x-y)^2$: the identity for x^2 is this variance.

3 Gradient descent

An iterative method starts from x_0 and repeats

$$d_k = \mathcal{D}(x_k, f, \nabla f, \dots), \qquad \alpha_k = \mathcal{S}(x_k, d_k), \qquad x_{k+1} = x_k + \alpha_k d_k.$$

Three design questions define a method: the direction d_k, the step α_k and the rule for stopping. Gradient descent takes $d_k = -\nabla f(x_k)$. This section treats a constant step; backtracking and stopping come in the next one.

> **Definition: Gradient descent.** Let $f : \mathbb{R}^n \to \mathbb{R}$ be differentiable and $\alpha > 0$. Gradient descent with constant step α from x_0 is the sequence
>
> $$x_{k+1} = x_k - \alpha \nabla f(x_k), \qquad k = 0, 1, 2, \dots$$
>
> Each iteration costs one gradient evaluation.

3.1 Why the negative gradient

> **Theorem: Steepest ascent in the Euclidean norm.** Let f be differentiable at x with $\nabla f(x) \neq 0$. Then
>
> $$\arg\max_{\|d\|_2 \le 1} \lim_{t \downarrow 0} \frac{f(x+td) - f(x)}{t} = \frac{\nabla f(x)}{\|\nabla f(x)\|_2},$$
>
> and the maximal rate of increase is $\|\nabla f(x)\|_2$.

Proof. Differentiability gives $f(x+td) = f(x)+t\,\nabla f(x)^\top d+o(t)$, so the limit equals $\nabla f(x)^\top d$. By the Cauchy–Schwarz inequality

$$\nabla f(x)^\top d \le \|\nabla f(x)\|_2\,\|d\|_2 \le \|\nabla f(x)\|_2,$$

with equality for $d = \nabla f(x)/\|\nabla f(x)\|_2$. Equality in Cauchy–Schwarz forces d to be a nonnegative multiple of $\nabla f(x)$, and equality in the second step forces $\|d\|_2 = 1$, so this maximizer is unique.

Replacing d by $-d$, the steepest descent direction among unit vectors is $-\nabla f(x)/\|\nabla f(x)\|_2$, with rate $-\|\nabla f(x)\|_2$.

The word *steepest* refers to the Euclidean unit ball. For a general norm the steepest unit direction solves $\max_{\|d\|\le 1} \nabla f(x)^\top d = \|\nabla f(x)\|_*$, where $\|\cdot\|_*$ is the dual norm, and the answer changes with the norm. For $\|d\|_M = \sqrt{d^\top M d}$ with $M \succ 0$ substitute $u = M^{1/2}d$: the problem becomes $\max_{\|u\|_2\le 1} \big(M^{-1/2}\nabla f(x)\big)^\top u$, the theorem gives $u \propto M^{-1/2}\nabla f(x)$, and so $d \propto M^{-1}\nabla f(x)$. The steepest descent direction in this norm points along $-M^{-1}\nabla f(x)$, the idea behind preconditioning and the natural gradient. For $\|\cdot\|_\infty$ one steepest descent direction is $-\operatorname{sign}\nabla f(x)$, taken componentwise.

3.2 The descent lemma and a safe step

Lemma: Descent lemma. Let $f : \mathbb{R}^n \to \mathbb{R}$ be differentiable with an L-Lipschitz gradient. Then for all x, y

$$f(y) \le f(x) + \nabla f(x)^\top (y-x) + \frac{L}{2}\|y-x\|^2.$$

Proof. Put $r(t) = f\big(x + t(y-x)\big)$ for $t \in [0,1]$. It is continuously differentiable with $r'(t) = \nabla f\big(x+t(y-x)\big)^\top (y-x)$, so

$$f(y) - f(x) = \int_0^1 r'(t)\,dt = \nabla f(x)^\top (y-x) + \int_0^1 \big(\nabla f(x+t(y-x)) - \nabla f(x)\big)^\top (y-x)\,dt.$$

By Cauchy–Schwarz and the Lipschitz condition the integrand is at most $\|\nabla f(x+t(y-x)) - \nabla f(x)\|\,\|y-x\| \le Lt\|y-x\|^2$, and $\int_0^1 Lt\,dt = \frac{L}{2}$.

The same estimate with absolute values gives $\big|f(y) - f(x) - \nabla f(x)^\top (y-x)\big| \le \frac{L}{2}\|y-x\|^2$: an L-smooth function stays between two parabolas around its tangent plane. Gradient descent needs only the upper one. The statement is global: one constant L controls the error of the linear model at every point and for every step length.

Corollary: One-step decrease. Let f be L-smooth, $\alpha > 0$ and $x^+ = x - \alpha\nabla f(x)$. Then

$$f(x^+) \le f(x) - \alpha\Big(1 - \frac{L\alpha}{2}\Big)\|\nabla f(x)\|^2.$$

Proof. Put $y = x^+$ in the descent lemma. Since $y - x = -\alpha\nabla f(x)$,

$$f(x^+) \le f(x) - \alpha\|\nabla f(x)\|^2 + \frac{L\alpha^2}{2}\|\nabla f(x)\|^2.$$

Two constants follow. For $0 < \alpha < 2/L$ the factor $1 - L\alpha/2$ is positive, so every step from a non-

stationary point strictly decreases f. The guaranteed decrease $\alpha(1 - L\alpha/2)$ is a concave parabola in α with its maximum at $\alpha = 1/L$, where

$$f(x^+) \le f(x) - \frac{1}{2L}\|\nabla f(x)\|^2,$$

the best guarantee this bound can give. Every $\alpha \le 1/L$ has $1 - L\alpha/2 \ge \frac{1}{2}$, hence $f(x^+) \le f(x) - \frac{\alpha}{2}\|\nabla f(x)\|^2$.

> **Trap.** The value $2/L$ is where the guarantee of strict decrease ends, not a proof that larger steps diverge: at $\alpha = 2/L$ the bound only rules out an increase, for $\alpha > 2/L$ it allows one, and a particular function may still decrease. For a quadratic, $2/L$ is the exact boundary, as the next subsection shows.

3.3 Quadratics and the exact stability condition

Take $f(x) = \frac{1}{2}x^\top Qx - b^\top x$ with symmetric $Q \succ 0$. Then $\nabla f(x) = Qx - b$, the unique minimizer is $x^* = Q^{-1}b$, and $\nabla f(x) = Q(x - x^*)$. For the error $e_k = x_k - x^*$ one step of gradient descent gives

$$e_{k+1} = e_k - \alpha Q e_k = (I - \alpha Q)\, e_k.$$

Diagonalize $Q = U\Lambda U^\top$ with U orthogonal and $\Lambda = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$, and pass to the eigen-coordinates $z_k = U^\top e_k$. The coordinates decouple:

$$z_{k+1,i} = (1 - \alpha\lambda_i)\, z_{k,i}, \qquad z_{k,i} = (1 - \alpha\lambda_i)^k z_{0,i},$$

and $f(x_k) - f^* = \frac{1}{2}e_k^\top Q e_k = \frac{1}{2}\sum_i \lambda_i z_{k,i}^2$.

> **Theorem: Stability on a quadratic.** Let $f(x) = \frac{1}{2}x^\top Qx - b^\top x$ with symmetric $Q \succ 0$. Gradient descent with a constant step α converges to x^* from every starting point if and only if
>
> $$0 < \alpha < \frac{2}{\lambda_{\max}(Q)}.$$

> ***Proof.*** By the recursion, $z_{k,i} \to 0$ for every $z_{0,i}$ if and only if $|1 - \alpha\lambda_i| < 1$, and $\|x_k - x^*\| = \|z_k\|$ because U is orthogonal. So $x_k \to x^*$ from every x_0 if and only if $|1 - \alpha\lambda_i| < 1$ for every i. Since $\lambda_i > 0$, this means $0 < \alpha\lambda_i < 2$ for all i, and the binding eigenvalue is the largest.

At the boundary $\alpha = 2/\lambda_{\max}$ the factor of the stiffest coordinate is -1: that coordinate flips sign with constant magnitude forever, while the others still shrink. Beyond the boundary it grows geometrically. Between $1/\lambda_{\max}$ and $2/\lambda_{\max}$ its factor is negative but less than 1 in magnitude, and it converges with alternating signs. Here $L = \lambda_{\max}(Q)$, so the safe range $0 < \alpha < 2/L$ from the descent lemma is exactly the stability range: for quadratics the lemma is sharp.

> **Example.** In A1, $Q = \operatorname{diag}(1,\ 10)$, so $L = 10$ and the boundary is $2/L = 0.2$. At $\alpha = 0.2$ the factors are 0.8 and -1. From $x_0 = (2,\ 2)$ the stiff coordinate alternates between 2 and -2, and
>
> $$f(x_k) = 20 + 2 \cdot 0.64^k :$$
>
> the values decrease towards 20 and never go below it, because the stiff coordinate alone contributes $\frac{1}{2} \cdot 10 \cdot 2^2 = 20$.

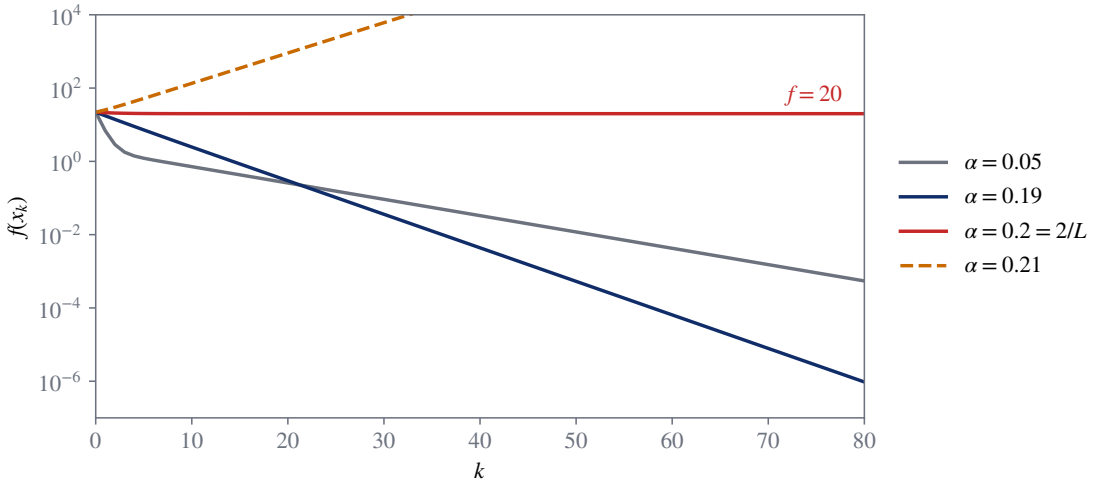


Figure. Values $f(x_k)$, log scale, for $f(x) = \frac{1}{2}x^\top Qx$ with $Q = \mathrm{diag}(1,\ 10)$, $x_0 = (2,\ 2)$ and the four steps of A1. Below the boundary $2/L = 0.2$ the values fall geometrically, at $\alpha = 2/L$ they level off at 20, and above it they grow.

3.4 Conditioning and the optimal constant step

For the quadratic, let $\mu = \lambda_{\min}(Q)$ and $L = \lambda_{\max}(Q)$. Per step the norm of the error is multiplied by at most

$$\rho(\alpha) = \max_i |1 - \alpha\lambda_i| = \max\big(|1 - \alpha\mu|,\ |1 - \alpha L|\big),$$

because $|1-\alpha\lambda|$ is convex in λ and its maximum over $[\mu, L]$ sits at an endpoint. With $\|e_k\| = \|z_k\|$ this gives

$$\|x_k - x^*\| \le \rho(\alpha)^k \, \|x_0 - x^*\|, \qquad f(x_k) - f^* \le \rho(\alpha)^{2k}\big(f(x_0) - f^*\big).$$

At $\alpha = 1/L$ the factors $1 - \lambda_i/L$ lie in $[0, 1 - 1/\kappa]$, so $\rho(1/L) = 1 - 1/\kappa$. Since $\ln \frac{1}{1-1/\kappa} \ge \frac{1}{\kappa}$, the error is below $\varepsilon\|x_0 - x^*\|$ once $k \ge \kappa \ln(1/\varepsilon)$: the iteration count grows in proportion to κ. Geometrically, a large κ is a long narrow valley; the gradient points mostly across it, and a step short enough for the steep direction makes little progress along the flat one. On a quadratic the gap therefore shrinks by at least $(1 - \mu/L)^2$ per step at $\alpha = 1/L$. This is sharper than the general guarantee $f(x_k) - f^* \le (1 - \mu/L)^k\big(f(x_0) - f^*\big)$, which holds at $\alpha = 1/L$ for every μ-strongly convex L-smooth function and is proved in the section on rates. A2 and A3 ask for that general guarantee, so their factor is $1 - \mu/L$, not $(1 - \mu/L)^2$.

Proposition: Optimal constant step. For a quadratic with $\mu = \lambda_{\min}(Q) > 0$ and $L = \lambda_{\max}(Q)$, the factor $\rho(\alpha)$ is minimized over $\alpha > 0$ by

$$\alpha^* = \frac{2}{L + \mu}, \qquad \rho(\alpha^*) = \frac{L - \mu}{L + \mu} = \frac{\kappa - 1}{\kappa + 1}.$$

Proof. For $\alpha > 0$, $|1 - \alpha\lambda| = \max(1 - \alpha\lambda,\ \alpha\lambda - 1)$; the first term is largest at $\lambda = \mu$ and the second at $\lambda = L$. So $\rho(\alpha) = \max(1 - \alpha\mu,\ \alpha L - 1)$, the maximum of a decreasing and an increasing linear function of α. It is smallest where the two are equal: $1 - \alpha\mu = \alpha L - 1$ gives $\alpha^* = 2/(L + \mu)$ and $\rho(\alpha^*) = 1 - \frac{2\mu}{L+\mu} = \frac{L-\mu}{L+\mu}$. For $\alpha < \alpha^*$ the first function is larger than at α^*, and for $\alpha > \alpha^*$ the second is, so no other step does as well. Dividing by μ gives $(\kappa - 1)/(\kappa + 1)$.

At α^* the two extreme factors are $1 - \alpha^*\mu = \rho(\alpha^*)$ and $1 - \alpha^* L = -\rho(\alpha^*)$, equal in magnitude. In A3, $\mu = 1$ and $L = 50$ give $\alpha^* = \frac{2}{51} \approx 0.039216$, and $\kappa = 50$ gives $\rho(\alpha^*) = \frac{49}{51} \approx 0.960784$,

against $1 - 1/\kappa = 0.98$ at $\alpha = 1/L$. When Q has only the eigenvalues μ and L, as in every two-dimensional example of the seminar, each coordinate shrinks by exactly $\rho(\alpha^*)$ in magnitude, and $f(x_k) - f^* = \rho(\alpha^*)^{2k}\big(f(x_0) - f^*\big)$ holds with equality. The same step and factor bound $\|x_k - x^*\|$ for every μ-strongly convex L-smooth function, with a longer proof.

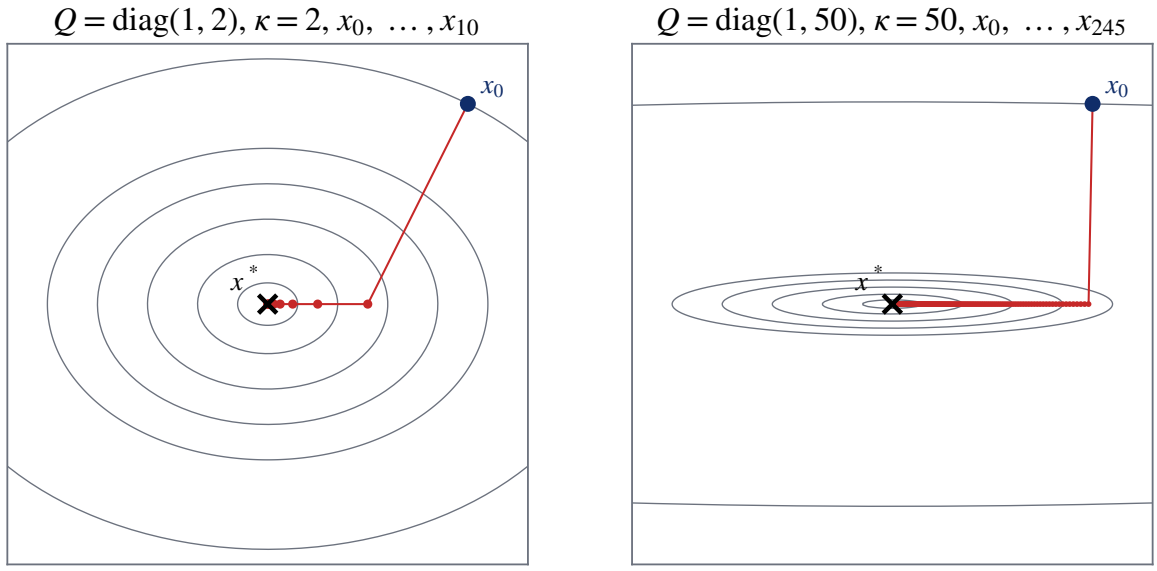


Figure. Contours of $f(x) = \frac{1}{2}x^\top Qx$ and the iterates of gradient descent with $\alpha = 1/L$ from $x_0 = (1,\ 1)$. Left: $Q = \mathrm{diag}(1,\ 2)$, $\kappa = 2$. Right: $Q = \mathrm{diag}(1,\ 50)$, $\kappa = 50$. The flat coordinate shrinks by $1 - 1/\kappa$ per step, 0.5 on the left and 0.98 on the right, and reaching $f(x_k) \le 10^{-6}\, f(x_0)$ takes 10 iterations on the left and 245 on the right.

3.5 Seminar problems

The problems of block A with complete solutions.

A1. Four steps on one quadratic

On the board

Statement. Minimize $f(x) = \frac{1}{2}x^\top Qx$ with $Q = \mathrm{diag}(1,\ 10)$ by gradient descent with a constant step α from $x_0 = (2,\ 2)$. **(a)** Find the stability boundary: the value $\alpha_{\max}$ such that the method converges from every start if and only if $0 < \alpha < \alpha_{\max}$. **(b)** Classify the steps $\alpha = 0.05,\ 0.19,\ 0.2$ and 0.21 through the per-coordinate recurrence $z_{k+1,i} = (1 - \alpha\lambda_i)\, z_{k,i}$: does the method converge, and how?

Solution.

1. Here $x^* = 0$, $f^* = 0$, and Q is already diagonal, so the eigen-coordinates are the coordinates: $z_k = x_k$, $\mu = \lambda_1 = 1$, $L = \lambda_2 = 10$, $\kappa = 10$. The step $x_{k+1} = x_k - \alpha Qx_k$ splits into $z_{k+1,i} = (1 - \alpha\lambda_i)\, z_{k,i}$, and $f(x_k) = \frac{1}{2}\big(\lambda_1 z_{k,1}^2 + \lambda_2 z_{k,2}^2\big)$, so $f(x_0) = 22$.
2. Convergence from every start needs $|1 - \alpha\lambda_i| < 1$ for both i, that is $0 < \alpha < 2/\lambda_i$. The flat coordinate allows $\alpha < 2/\mu = 2$, the stiff one only $\alpha < 2/L = 0.2$. The stiff direction sets the boundary: $0 < \alpha < 0.2$.

3. The factors of the four steps:

α	$1-\alpha\lambda_1$	$1-\alpha\lambda_2$
0.05	0.95	0.5
0.19	0.81	-0.9
0.2	0.8	-1
0.21	0.79	-1.1

4. $\alpha = 0.05$: both factors lie in $(0, 1)$, so both coordinates shrink without changing sign. The stiff one halves at every step, the flat one loses only a twentieth. After ten steps $f = 0.7170$, and almost all of it sits in the flat coordinate. Converges, slowly.

5. $\alpha = 0.19$: the stiff factor -0.9 is negative, so $z_{k,2}$ changes sign at every step, but its magnitude still shrinks. Converges with sign flips. Early on the stiff coordinate dominates: it holds 20 of $f(x_0) = 22$ and shrinks by $|-0.9|$ here against 0.5 at $\alpha = 0.05$, so this step is behind after ten iterations, $f = 2.4611$ against 0.7170. In the long run the pace is set by the larger factor in magnitude, $|-0.9|$ here against 0.95, so it is far ahead after fifty, $f = 5.31 \cdot 10^{-4}$ against $1.18 \cdot 10^{-2}$.

6. $\alpha = 0.2 = 2/L$: the stiff factor is exactly -1. The stiff coordinate flips between $+2$ and -2 forever and alone contributes $\frac{1}{2} \cdot 10 \cdot 2^2 = 20$ to f at every step. The flat coordinate still shrinks by 0.8, so $f(x_k) = 20 + 2 \cdot 0.64^k$: it decreases towards 20 and never goes below it (20.0231 after ten steps). Neither convergence to x^* nor divergence.

7. $\alpha = 0.21$: $|-1.1| > 1$, so $|z_{k,2}|$ grows at every step while its sign alternates, and $f(x_k) \to \infty$: 134.5679 after ten steps, $2.76 \cdot 10^5$ after fifty. The flat coordinate converges, which does not help. Diverges.

8. The boundary comes from the largest eigenvalue, the speed from the smallest: for every $\alpha < 2/L$ the flat factor is $1 - \alpha\mu > 1 - 2/\kappa$. A large κ forces a step that barely moves the flat coordinate.

Answer. Stable for $0 < \alpha < 2/L = 0.2$. $\alpha = 0.05$: converges, slowly in the flat coordinate; 0.19: converges with sign flips in the stiff coordinate; 0.2: the stiff coordinate flips between ± 2 forever and $f(x_k)$ decreases towards 20, never below; 0.21: diverges.

A2. Guaranteed and observed iterations

On your own

Statement. Run gradient descent on $f(x) = \frac{1}{2} x^\top Q x$ with $Q = \mathrm{diag}(1,\ 2)$, step $\alpha = 1/L$, from $x_0 = (1,\ 1)$. Report **(a)** κ; **(b)** the factor q in the lecture's guarantee $f(x_k) - f^* \le q^k \left(f(x_0) - f^*\right)$; **(c)** the smallest k for which this bound gives $f(x_k) - f^* \le 10^{-6} \left(f(x_0) - f^*\right)$; **(d)** the smallest k for which the iterates actually satisfy it.

Solution.

1. $L = \lambda_{\max} = 2$, $\mu = \lambda_{\min} = 1$, so $\kappa = 2$ and $\alpha = 1/L = 0.5$.

2. The linear rate at $\alpha = 1/L$ is $f(x_k) - f^* \le (1 - \mu/L)^k \left(f(x_0) - f^*\right)$, so $q = 1 - \mu/L = 0.5$.

3. The bound reaches 10^{-6} when $0.5^k \le 10^{-6}$, that is $k \ge \dfrac{\ln 10^{-6}}{\ln 0.5} = 19.93$. Round up: 20 iterations.

4. Now the run itself, with $f^* = 0$. The factors $1 - \alpha\lambda_i$ are 0.5 for $\lambda = 1$ and 0 for $\lambda = 2$: the stiff coordinate is gone after the first step.

5. So $f(x_0) = \frac{1}{2}(1 + 2) = 1.5$ and $f(x_k) = \frac{1}{2} \cdot 0.5^{2k}$ for $k \ge 1$. From then on the gap shrinks by 0.5^2 per iteration, not by 0.5.

6. $f(x_k) \le 10^{-6} f(x_0)$ means $0.5^{2k} \le 2 \cdot 1.5 \cdot 10^{-6}$, that is $k \ge \dfrac{\ln\left(2 \cdot 1.5 \cdot 10^{-6}\right)}{2 \ln 0.5} = 9.17$. The run needs 10 iterations.

7. The bound is weaker because its proof uses only the descent inequality and $\|\nabla f\|^2 \ge 2\mu\,(f - f^*)$, which hold for every function with these μ and L. On a quadratic at $\alpha = 1/L$ each coordinate term of the gap shrinks by $(1 - \lambda_i/L)^2 \le (1 - \mu/L)^2$.

Answer. $\kappa = 2$, $q = 0.5$, the bound allows 20 iterations, the run needs 10.

A3. An ill-conditioned bowl and the optimal step

On your own

Statement. Same setup as A2 with $Q = \mathrm{diag}(1,\ 50)$ and $x_0 = (1,\ 1)$. **(a)–(d)** Report the same four quantities at $\alpha = 1/L$. **(e)** Give the optimal constant step $\alpha^* = 2/(L + \mu)$ as a number. **(f)** With α^* every coordinate of the error $x_k - x^*$ is multiplied at each step by a factor of magnitude at most $\rho = (\kappa - 1)/(\kappa + 1)$. How many iterations does the resulting bound on the gap $f(x_k) - f^*$ need for the same reduction by 10^{-6}?

Solution.

1. $L = 50$, $\mu = 1$, $\kappa = 50$, $\alpha = 1/L = 0.02$ and $q = 1 - \mu/L = 0.98$.

2. Bound: $k \ge \dfrac{\ln 10^{-6}}{\ln 0.98} = 683.84$, so 684 iterations.

3. The factors at $\alpha = 1/L$ are $1 - \mu/L = 0.98$ and $1 - L/L = 0$. The stiff coordinate holds $\frac{1}{2} \cdot 50$ out of $f(x_0) = 25.5$ and is gone after one step; then $f(x_k) = \frac{1}{2} \cdot 0.98^{2k}$ for $k \ge 1$.

4. $f(x_k) \le 10^{-6} f(x_0)$ means $0.98^{2k} \le 2 \cdot 25.5 \cdot 10^{-6}$, that is $k \ge \dfrac{\ln\left(2 \cdot 25.5 \cdot 10^{-6}\right)}{2 \ln 0.98} = 244.6$. The run needs 245 iterations against the bound's 684.

5. Optimal step: $\alpha^* = \dfrac{2}{L + \mu} = \frac{2}{51} \approx 0.039216$. The factors are $1 - \alpha^*\mu = \frac{49}{51}$ and $1 - \alpha^* L = -\frac{49}{51}$: equal magnitudes, $\rho = \dfrac{\kappa - 1}{\kappa + 1} = \frac{49}{51} \approx 0.960784$.

6. The gap is quadratic in the coordinates, so $f(x_k) - f^* \le \rho^{2k}\left(f(x_0) - f^*\right)$. The bound needs $k \ge \dfrac{\ln 10^{-6}}{2 \ln \rho} = 172.67$, so 173 iterations. With ρ^k instead of ρ^{2k} one gets 346, the count for the distance $\|x_k - x^*\|$.

7. Here both coordinates shrink by exactly ρ in magnitude, so $f(x_k) = \rho^{2k} f(x_0)$ with equality, and the run also needs 173 iterations. With α^* the bound and the run coincide; at $\alpha = 1/L$ they differ almost threefold.

Answer. $\kappa = 50$, $q = 0.98$, the bound allows 684 iterations, the run needs 245; $\alpha^* = \frac{2}{51} \approx 0.039216$, and the bound at α^* allows 173, exactly what the run needs.

4 Line search and stopping

A constant step needs L, which is rarely known, and a step that is safe where f curves most is too short elsewhere. Backtracking chooses the step at every iteration from values of f along the ray.

4.1 Armijo backtracking

Definition: Armijo backtracking. Fix an initial step $\alpha_0 > 0$, a shrinkage factor $\beta \in (0,1)$ and a constant $c \in (0,1)$, and let d be a descent direction at x. Start with $\alpha = \alpha_0$ and repeat $\alpha \leftarrow \beta\alpha$ until the sufficient decrease (Armijo) condition

$$f(x + \alpha d) \le f(x) + c\,\alpha\,\nabla f(x)^{\top} d$$

holds. The accepted step is the first $\alpha = \beta^j \alpha_0$, $j = 0, 1, 2, \dots$, that passes. For gradient descent $d = -\nabla f(x)$ and the right-hand side is $f(x) - c\,\alpha \|\nabla f(x)\|^2$. Typical values are $\beta = \frac{1}{2}$ and $c = 10^{-4}$.

Along the ray put $\varphi(\alpha) = f(x + \alpha d)$, so $\varphi(0) = f(x)$ and $\varphi'(0) = \nabla f(x)^{\top} d < 0$. The Armijo line $\varphi(0) + c\alpha\varphi'(0)$ is the tangent line with its slope reduced by the factor c, so for $\alpha > 0$ it lies above the tangent. Since $\varphi(\alpha) = \varphi(0) + \alpha\varphi'(0) + o(\alpha)$, short steps pass; a long step fails when the curvature bends φ back above the line. The test rejects steps whose decrease is a negligible fraction of what the slope promises.

Theorem: Termination of backtracking. Let f be L-smooth, $\nabla f(x) \neq 0$, $d = -\nabla f(x)$, $\alpha_0 > 0$ and $\beta, c \in (0,1)$. The Armijo condition holds for every $\alpha \in \big(0, 2(1-c)/L\big]$. Consequently backtracking stops after finitely many reductions, and the accepted step satisfies

$$\alpha \ge \min\Big(\alpha_0,\ \frac{2\beta(1-c)}{L}\Big).$$

Proof. By the one-step decrease bound, $f\big(x - \alpha\nabla f(x)\big) \le f(x) - \alpha\big(1 - \frac{L\alpha}{2}\big)\|\nabla f(x)\|^2$. If $\alpha \le 2(1-c)/L$, then $1 - L\alpha/2 \ge c$ and the right-hand side is at most $f(x) - c\alpha\|\nabla f(x)\|^2$: the Armijo condition holds. The trial steps $\beta^j\alpha_0$ tend to 0, so a finite j brings them into $\big(0, 2(1-c)/L\big]$, and the loop stops there at the latest. If α_0 passes, the accepted step is α_0. Otherwise the accepted step α follows a rejected trial α/β, every rejected trial exceeds $2(1-c)/L$, and so $\alpha > 2\beta(1-c)/L$.

The proof also bounds the work: the number of reductions is at most the smallest $j \ge 0$ with $\beta^j\alpha_0 \le 2(1-c)/L$, that is $\big\lceil \log_{1/\beta}\big(\alpha_0 L/(2(1-c))\big)\big\rceil$ when $\alpha_0 > 2(1-c)/L$. The loop may stop earlier, because along a particular direction the condition can hold beyond $2(1-c)/L$. On a quadratic the range is exact: with $g = \nabla f(x)$,

$$f(x - \alpha g) = f(x) - \alpha\, g^{\top} g + \frac{\alpha^2}{2}\, g^{\top} Q g,$$

so the Armijo condition holds if and only if $\alpha \le 2(1-c)\,\dfrac{g^{\top} g}{g^{\top} Q g}$, and this is at least $2(1-c)/L$ because $g^{\top} Q g \le L\, g^{\top} g$.

Example. In B1, $Q = \mathrm{diag}(1,\ 10)$, $x_0 = (2,\ 2)$, $L = 10$ and $c = 10^{-4}$. The theorem guarantees acceptance for $\alpha \le 2(1-c)/L = 0.19998$, and along $d = -\nabla f(x_0)$ the exact range is $\alpha \le 0.20178$. Halving from $\alpha_0 = 1$, the first trial inside both ranges is 0.125, reached after

3 halvings; it lies between $1/L = 0.1$ and $2/L = 0.2$.

Without L-smoothness backtracking still terminates for any descent direction d of a differentiable f, because $f(x + \alpha d) - f(x) - c\alpha\nabla f(x)^\top d = (1 - c)\,\alpha\,\nabla f(x)^\top d + o(\alpha)$ is negative for small α. What is lost is the lower bound on the accepted step.

Trap. If d is not a descent direction, short trial steps fail the test and the loop may never end. An implementation checks $\nabla f(x)^\top d < 0$ and caps the number of reductions.

4.2 Stopping rules

The lecture uses three scaled tests with tolerances ε_g, ε_x and ε_f:

$$\|\nabla f(x_k)\| \le \varepsilon_g \max\big(1, |f(x_k)|\big),$$

$$\|x_{k+1} - x_k\| \le \varepsilon_x \max\big(1, \|x_k\|\big),$$

$$|f(x_{k+1}) - f(x_k)| \le \varepsilon_f \max\big(1, |f(x_k)|\big).$$

The factor $\max(1, \cdot)$ makes a test relative when the scale is large and absolute when it is below 1, so a quantity near zero is not asked for an impossible relative accuracy.

In code the tolerance is often split into an absolute and a relative part, $\text{atol} + \text{rtol} \cdot \max(1, s)$, where s is the scale, $|f(x_k)|$ or $\|x_k\|$. B3 uses this form with $\text{atol} = 10^{-8}$ and $\text{rtol} = 10^{-6}$.

The tests measure different things. The change tests watch the motion of the method: for gradient descent $\|x_{k+1} - x_k\| = \alpha_k\|\nabla f(x_k)\|$ and $f(x_k) - f(x_{k+1}) \approx \alpha_k\|\nabla f(x_k)\|^2$, and both are small whenever the step is small, however far the point is from a solution. Small changes can mean convergence, a tiny step, poor scaling or floating-point stagnation. The gradient test looks at the point itself. A small gradient is a first-order certificate of stationarity; under the PL condition it bounds the error in f, and for a μ-strongly convex f also the distance to x^*:

$$f(x) - f^* \le \frac{1}{2\mu}\|\nabla f(x)\|^2, \qquad \|x - x^*\| \le \frac{1}{\mu}\|\nabla f(x)\|.$$

The first is PL rearranged. For the second, add the first-order form of strong convexity at x and at x^* to get $\big(\nabla f(x) - \nabla f(x^*)\big)^\top (x - x^*) \ge \mu\|x - x^*\|^2$, then use $\nabla f(x^*) = 0$ and Cauchy–Schwarz.

Combined rule

Stop with success when the gradient test holds, with the tolerance scaled by $\max(1, |f(x_k)|)$ as in the lecture. Treat the change tests only as a stagnation signal: if they hold while the gradient test fails, stop and report that the method stalled, not that it converged. Always add a cap on iterations and function evaluations, and stop when f or ∇f is not finite.

4.3 A change-based test that stops too early

Problem B3 builds the failure on purpose. Take $f(x) = \frac{1}{2}x^\top Q x$ with $Q = \text{diag}(1,\ 50)$, so that $x^* = 0$ and $f^* = 0$, start at $x_0 = (1,\ 1)$ with $f(x_0) = 25.5$, and make one gradient step with the prescribed step $\alpha = 10^{-8}$. It gives $x_1 = (0.99999999,\ 0.9999995)$ and $f(x_1) = 25.49997499$. The three tests with $\text{atol} = 10^{-8}$ and $\text{rtol} = 10^{-6}$ compare as follows.

Test	Value	Threshold	Fires
function change $\lvert f(x_1) - f(x_0)\rvert$	$2.501 \cdot 10^{-5}$	$\text{atol} + \text{rtol}\max(1, \lvert f(x_0)\rvert) = 2.551 \cdot 10^{-5}$	yes
step $\lVert x_1 - x_0\rVert$	$5.001 \cdot 10^{-7}$	$\text{atol} + \text{rtol}\max(1, \lVert x_0\rVert) = 1.424 \cdot 10^{-6}$	yes
gradient $\lVert \nabla f(x_1)\rVert$	50.01	$\text{atol} + \text{rtol}\max(1, \lvert f(x_1)\rvert) = 2.551 \cdot 10^{-5}$	no

Both change tests fire after one step, although f has dropped by less than a millionth of its value and the point has hardly moved towards x^*. Only the gradient test refuses to stop. The change tests fire because their left-hand sides are proportional to the step: $\lVert x_1 - x_0\rVert = \alpha\lVert \nabla f(x_0)\rVert$ and $\lvert f(x_1) - f(x_0)\rvert \approx \alpha\lVert \nabla f(x_0)\rVert^2$ with $\lVert \nabla f(x_0)\rVert = 50.01$, and a tiny step makes both tiny. The gradient test has no factor α: it measures the gradient at x_1 itself, and a tiny step leaves it almost as large as at x_0, $\lVert \nabla f(x_1)\rVert \approx \lVert \nabla f(x_0)\rVert$.

4.4 Seminar problems

The problems of block B with complete solutions.

B1. Armijo backtracking by hand

On the board

Statement. Let $f(x) = \frac{1}{2}x^\top Q x$ with $Q = \operatorname{diag}(1, 10)$ and $x_0 = (2,\ 2)$. Run Armijo backtracking along $d = -\nabla f(x_0)$ with $\alpha_0 = 1$, $\beta = 0.5$ and $c = 10^{-4}$: try $\alpha = \alpha_0$ first and set $\alpha \leftarrow \beta\alpha$ while $f(x_0 + \alpha d) > f(x_0) + c\alpha \nabla f(x_0)^\top d$. How many halvings happen before the test passes, which step is accepted, and how does it compare with $1/L$ and $2/L$?

Solution.

1. Data at x_0: $f(x_0) = 22$, $\nabla f(x_0) = Qx_0 = (2,\ 20)$ and $\nabla f(x_0)^\top d = -\lVert \nabla f(x_0)\rVert^2 = -404$. Here $L = 10$, $1/L = 0.1$ and $2/L = 0.2$.

2. Along the ray each coordinate gets its own factor: $x_0 + \alpha d = \big(2(1-\alpha),\ 2(1-10\alpha)\big)$. The test reads $f(x_0 + \alpha d) \le 22 - 10^{-4} \cdot 404\,\alpha$.

3. Trials, halving from α_0:

α	$f(x_0 + \alpha d)$	$f(x_0) + c\alpha\nabla f(x_0)^\top d$	passes
1	1620	21.9596	no
0.5	320.5	21.9798	no
0.25	46.125	21.9899	no
0.125	2.78125	21.99495	yes

4. The first trial that passes is $\alpha = 0.125$, after 3 halvings, and the loop stops there. It lies between $1/L = 0.1$ and $2/L = 0.2$, inside the stable range of block A.

5. Why the loop must stop: the descent lemma gives $f(x_0 + \alpha d) \le f(x_0) - \alpha(1 - L\alpha/2)\lVert \nabla f(x_0)\rVert^2$, so the test holds for every $\alpha \le 2(1-c)/L = 0.19998$. Every rejected trial is above this value, so the accepted one is at least β times it.

6. On a quadratic the test can be solved exactly. With $g = \nabla f(x_0)$, $f(x_0 - \alpha g) = f(x_0) - \alpha\lVert g\rVert^2 + \frac{1}{2}\alpha^2 g^\top Q g$, so the test holds for $\alpha \le 2(1-c)\lVert g\rVert^2/(g^\top Q g) = 0.20178$, a little above 0.19998 because the curvature along g is below L. The minimizer along the ray is $\lVert g\rVert^2/(g^\top Q g) = 0.1009$; Armijo does not look for it.

Answer. 3 halvings, accepted $\alpha = 0.125$, between $1/L = 0.1$ and $2/L = 0.2$; every $\alpha \le 2(1-c)/L = 0.19998$ passes.

B2. Backtracking on a stiffer quadratic

On your own

Statement. Let $f(x) = \frac{1}{2}x^\top Qx$ with $Q = \mathrm{diag}(1, 50)$ and $x_0 = (2,\ 2)$. Run the same backtracking as in B1 along $d = -\nabla f(x_0)$ with $\alpha_0 = 1$, $\beta = 0.5$ and $c = 10^{-4}$: try $\alpha = \alpha_0$ first and set $\alpha \leftarrow \beta\alpha$ while $f(x_0 + \alpha d) > f(x_0) + c\alpha\nabla f(x_0)^\top d$. How many halvings happen, which step is accepted, and where does it sit relative to $1/L$ and $2/L$?

Solution.

1. Data: $f(x_0) = 102$, $\nabla f(x_0) = (2,\ 100)$, $\|\nabla f(x_0)\|^2 = 10004$; $L = 50$, $1/L = 0.02$, $2/L = 0.04$. With $x_0 + \alpha d = \big(2(1-\alpha),\ 2(1-50\alpha)\big)$ the test is $f(x_0 + \alpha d) \le 102 - 10^{-4} \cdot 10004\,\alpha$.

2. Trials, halving from α_0:

α	$f(x_0 + \alpha d)$	$f(x_0) + c\alpha\nabla f(x_0)^\top d$	passes
1	240100	100.9996	no
0.5	57600.5	101.4998	no
0.25	13226.125	101.7499	no
0.125	2757.78125	101.87495	no
0.0625	453.3203125	101.937475	no
0.03125	33.517578125	101.9687375	yes

3. The first pass is $\alpha = 0.03125$ after 5 halvings. It lies between $1/L = 0.02$ and $2/L = 0.04$, as in B1.

4. Guarantee: every $\alpha \le 2(1-c)/L = 0.039996$ passes the test, and this bound is just below $2/L$. The halvings cross it between 0.0625 and 0.03125, and the accepted step is at least β times it.

5. Exact threshold on this quadratic: $2(1-c)\|g\|^2/(g^\top Qg) = 0.040012$ with $g = \nabla f(x_0)$, slightly above $2/L$. Armijo checks the decrease of one step, not stability: a trial just above $2/L$ would pass here, although as a constant step it diverges.

6. Compared with B1, L went from 10 to 50 and backtracking needed 5 halvings instead of 3. In both problems the accepted step lands between $1/L$ and $2/L$, and backtracking never used L.

Answer. 5 halvings, accepted $\alpha = 0.03125$, between $1/L = 0.02$ and $2/L = 0.04$ and below the guarantee $2(1-c)/L = 0.039996$.

B3. Three stopping tests after a tiny step

On your own

Statement. Let $f(x) = \frac{1}{2}x^\top Qx$ with $Q = \mathrm{diag}(1, 50)$ and $x_0 = (1,\ 1)$. Make one gradient step $x_1 = x_0 - \alpha\nabla f(x_0)$ with the prescribed step $\alpha = 10^{-8}$. With $\mathrm{atol} = 10^{-8}$, $\mathrm{rtol} = 10^{-6}$ and Euclidean norms, evaluate the three scaled tests

$$\begin{aligned} \text{gradient:} &\quad \|\nabla f(x_1)\| \le \mathrm{atol} + \mathrm{rtol} \cdot \max(1, |f(x_1)|), \\ \text{step:} &\quad \|x_1 - x_0\| \le \mathrm{atol} + \mathrm{rtol} \cdot \max(1, \|x_0\|), \\ \text{function:} &\quad |f(x_1) - f(x_0)| \le \mathrm{atol} + \mathrm{rtol} \cdot \max(1, |f(x_0)|). \end{aligned}$$

A test fires when its inequality holds. Report $|f(x_1) - f(x_0)|$, $\|\nabla f(x_1)\|$ and the number of tests that fire. Is x_1 close to the minimizer? Then state a combined stopping rule and say which quantity it is scaled by.

Solution.

1. One step: $\nabla f(x_0) = Qx_0 = (1, 50)$, $\|\nabla f(x_0)\| = 50.01$, and $x_1 = x_0 - \alpha \nabla f(x_0) = (0.99999999,\ 0.9999995)$.
2. Function test: $f(x_0) = 25.5$ and $f(x_1) = 25.49997499$, so $|f(x_1) - f(x_0)| = 2.501 \cdot 10^{-5}$, which matches the first-order value $\alpha\|\nabla f(x_0)\|^2$. The threshold is $\text{atol} + \text{rtol} \cdot 25.5 = 2.551 \cdot 10^{-5}$: the test fires.
3. Step test: $\|x_1 - x_0\| = \alpha\|\nabla f(x_0)\| = 5.001 \cdot 10^{-7}$ against $\text{atol} + \text{rtol} \cdot \max(1, \|x_0\|) = \text{atol} + \text{rtol}\sqrt{2} = 1.424 \cdot 10^{-6}$: it fires too.
4. Gradient test: $\|\nabla f(x_1)\| = 50.01$, almost the same as at x_0, against $\text{atol}+\text{rtol}\cdot\max(1, |f(x_1)|) = 2.551 \cdot 10^{-5}$. It does not fire, so 2 tests out of 3 fire.
5. The minimizer is $x^* = 0$ with $f^* = 0$, while $\|x_1 - x^*\| \approx \sqrt{2}$ and $f(x_1) \approx f(x_0)$. The change tests fired because both changes shrink with α (to first order they are proportional to it): they measure the step, not the distance to x^*.
6. Combined rule: report convergence only when $\|\nabla f(x_k)\| \le \text{atol} + \text{rtol} \cdot \max(1, |f(x_k)|)$, a test scaled by $\max(1, |f(x_k)|)$. If only the change tests fire, stop with a stagnation warning, not a solution; always cap the number of iterations and stop on non-finite values.

Answer. $|f(x_1) - f(x_0)| = 2.501 \cdot 10^{-5}$ and $\|x_1 - x_0\| = 5.001 \cdot 10^{-7}$ are below their thresholds, $\|\nabla f(x_1)\| = 50.01$ is not: 2 tests fire far from $x^* = 0$. Stop on the gradient test scaled by $\max(1, |f(x_k)|)$ and use the change tests only as a stagnation signal, with an iteration cap.

5 Rates

The lecture proves three guarantees for gradient descent with the step $\alpha = 1/L$: a gap of order $1/k$ for convex functions, and a gap that shrinks by the factor $1 - \mu/L$ at every step for strongly convex functions and, more generally, under the PL condition. All three proofs start from the descent step of the section on gradient descent: for an L-smooth f and $x_{k+1} = x_k - \frac{1}{L}\nabla f(x_k)$ the descent lemma gives

$$f(x_{k+1}) \le f(x_k) - \frac{1}{2L}\|\nabla f(x_k)\|^2.$$

In particular $f(x_k)$ never increases. Throughout, $f^* = \inf f$, and x^* is a minimizer when one exists.

5.1 The convex case

Theorem: Rate for convex functions. Let f be convex and L-smooth, and let x^* be a minimizer, $f^* = f(x^*)$. Gradient descent with $\alpha = 1/L$ satisfies

$$f(x_k) - f^* \le \frac{L\|x_0 - x^*\|^2}{2k} \qquad \text{for every } k \ge 1.$$

Proof. Write $g_k = \nabla f(x_k)$. Convexity puts the tangent plane at x_k below the graph, so $f^* \ge f(x_k) + g_k^\top(x^* - x_k)$, that is, $f(x_k) \le f^* + g_k^\top(x_k - x^*)$. Substituting this into the descent

step gives

$$f(x_{k+1}) - f^* \le g_k^\top (x_k - x^*) - \frac{1}{2L}\|g_k\|^2.$$

The right-hand side is a difference of squared distances. Since $x_{k+1} - x^* = (x_k - x^*) - \frac{1}{L} g_k$, expanding the square gives the three-point identity

$$\|x_{k+1} - x^*\|^2 = \|x_k - x^*\|^2 - \frac{2}{L}\, g_k^\top (x_k - x^*) + \frac{1}{L^2}\|g_k\|^2,$$

and multiplying it by $L/2$ turns the previous inequality into the key inequality

$$f(x_{k+1}) - f^* \le \frac{L}{2}\Big(\|x_k - x^*\|^2 - \|x_{k+1} - x^*\|^2\Big).$$

Summing over the steps $0, 1, \dots, k-1$, the right-hand side telescopes:

$$\sum_{j=1}^{k} \big(f(x_j) - f^*\big) \le \frac{L}{2}\Big(\|x_0 - x^*\|^2 - \|x_k - x^*\|^2\Big) \le \frac{L}{2}\|x_0 - x^*\|^2.$$

Finally, $f(x_j)$ does not increase, so each of the k terms on the left is at least $f(x_k) - f^*$. Hence $k\big(f(x_k) - f^*\big) \le \frac{L}{2}\|x_0 - x^*\|^2$.

The key inequality has a second consequence. Its left-hand side is nonnegative, so $\|x_{k+1} - x^*\| \le \|x_k - x^*\|$: the iterates never move away from any minimizer.

> **Why.** Each step pays for its gap with a decrease of the squared distance to x^*, and the whole budget is $\|x_0 - x^*\|^2$. So the gaps are small on average over the first k steps, and since they only decrease, the last one is at most the average.

The rate is **sublinear**: halving the bound takes twice as many steps, and a gap below ε is guaranteed only after $L\|x_0 - x^*\|^2/(2\varepsilon)$ steps.

> **Check yourself.** The theorem holds for every minimizer x^*. If the minimizers form a whole set, which of them gives the best bound?
>
> **Answer.** The one closest to x_0. All minimizers have the same value f^*, and the proof uses nothing else about x^*, so $\|x_0 - x^*\|$ can be replaced by the distance from x_0 to the set of minimizers.

5.2 Strong convexity

A differentiable f is μ-**strongly convex**, $\mu > 0$, if

$$f(y) \ge f(x) + \nabla f(x)^\top (y - x) + \frac{\mu}{2}\|y - x\|^2 \qquad \text{for all } x, y.$$

For an L-smooth f the descent lemma adds the upper bound, which sandwiches $f(y) - f(x) - \nabla f(x)^\top (y - x)$ between $\frac{\mu}{2}\|y - x\|^2$ and $\frac{L}{2}\|y - x\|^2$, so $\mu \le L$ and $\kappa = L/\mu \ge 1$.

> **Lemma: Strong convexity implies PL.** If f is μ-strongly convex, then for every x
>
> $$\frac{1}{2}\|\nabla f(x)\|^2 \ge \mu\big(f(x) - f^*\big).$$

Proof. Fix x and write $g = \nabla f(x)$. The right-hand side of the definition is a quadratic function of y,

$$q(y) = f(x) + g^\top (y - x) + \frac{\mu}{2}\|y - x\|^2.$$

Its gradient $g + \mu(y - x)$ vanishes at $y = x - g/\mu$, so its smallest value is

$$q\Big(x - \frac{g}{\mu}\Big) = f(x) - \frac{1}{\mu}\|g\|^2 + \frac{1}{2\mu}\|g\|^2 = f(x) - \frac{1}{2\mu}\|g\|^2.$$

Hence $f(y) \geq q(y) \geq f(x) - \frac{1}{2\mu}\|g\|^2$ for every y. Taking the infimum over y gives $f^* \geq f(x) - \frac{1}{2\mu}\|g\|^2$, and multiplying by μ and rearranging gives the claim.

Theorem: Linear rate under strong convexity. If f is μ-strongly convex and L-smooth, gradient descent with $\alpha = 1/L$ satisfies

$$f(x_{k+1}) - f^* \leq \Big(1 - \frac{\mu}{L}\Big)\big(f(x_k) - f^*\big), \qquad f(x_k) - f^* \leq \Big(1 - \frac{\mu}{L}\Big)^k \big(f(x_0) - f^*\big).$$

Proof. Subtract f^* from both sides of the descent step and bound the gradient term from below by the lemma, $\frac{1}{2L}\|\nabla f(x_k)\|^2 \geq \frac{\mu}{L}\big(f(x_k) - f^*\big)$:

$$f(x_{k+1}) - f^* \leq f(x_k) - f^* - \frac{1}{2L}\|\nabla f(x_k)\|^2 \leq \Big(1 - \frac{\mu}{L}\Big)\big(f(x_k) - f^*\big).$$

Applying this k times gives the second inequality.

The factor $1 - \mu/L = 1 - 1/\kappa$ lies in $[0, 1)$. A bound of the form Cq^k with $q < 1$ is called a **linear rate**: on a logarithmic scale the bound falls along a straight line.

Why. Strong convexity keeps the gradient large while the gap is large. With the step $1/L$ every iteration therefore removes at least the fraction μ/L of the remaining gap.

Check yourself. Take $f(x) = \frac{1}{2}x^\top Q x$ with $\kappa = 1$. What does the theorem give, and what does gradient descent with $\alpha = 1/L$ actually do?

Answer. The factor is $1 - \mu/L = 0$, so the gap vanishes after one step. Indeed $\kappa = 1$ means $Q = LI$, and $x_1 = x_0 - \frac{1}{L} \cdot L x_0 = 0 = x^*$.

5.3 The PL condition

The proof above used strong convexity only once, through the inequality of the lemma. That inequality is the PL condition, and it is enough on its own.

Theorem: Linear rate under PL. Let f be L-smooth with $f^* = \inf f > -\infty$, and let it satisfy the Polyak–Łojasiewicz condition with some $\mu > 0$:

$$\frac{1}{2}\|\nabla f(x)\|^2 \geq \mu\big(f(x) - f^*\big) \qquad \text{for all } x.$$

Then gradient descent with $\alpha = 1/L$ satisfies $f(x_k) - f^* \leq (1 - \mu/L)^k \big(f(x_0) - f^*\big)$.

Proof. The descent step gives $f(x_{k+1}) \le f(x_k) - \frac{1}{2L}\|\nabla f(x_k)\|^2$, and PL bounds the subtracted term from below by $\frac{\mu}{L}\big(f(x_k) - f^*\big)$. Subtracting f^* gives $f(x_{k+1}) - f^* \le (1 - \mu/L)\big(f(x_k) - f^*\big)$, and iterating gives the claim. Convexity was not used anywhere.

PL says that the gradient is large wherever the gap is large. At a stationary point the left-hand side is 0, so $f(x) = f^*$: **every stationary point is a global minimizer**. This is usually what convexity is needed for, and PL gives it without convexity.

Example. $f(x) = x^2 + 3\sin^2 x$ on $\mathbb{R}$ satisfies PL but is not convex. It is not convex because $f''(x) = 2 + 6\cos 2x$ equals $2 - 6 < 0$ at $x = \pi/2$. It is L-smooth with $L = 8$, since $|f''| \le 8$. Its only stationary point is the minimizer 0: for $0 < x < \frac{3}{2}$ both terms of $f'(x) = 2x + 3\sin 2x$ are positive, for $x \ge \frac{3}{2}$ we have $2x \ge 3 \ge -3\sin 2x$ with no equality in both places at once, and f' is odd. So the ratio $\frac{1}{2}f'(x)^2/f(x)$ is continuous and positive for $x \ne 0$. It tends to 8 as $x \to 0$, where $f(x) \approx 4x^2$ and $f'(x) \approx 8x$, and to 2 as $|x| \to \infty$. Its infimum is therefore positive, and PL holds with μ equal to it.

Trap. PL and convexity are independent. The example above is PL and not convex. The function $f(x) = x^4$ is convex and not PL: $\frac{1}{2}f'(x)^2 = 8x^6$ is smaller than μx^4 for every $\mu > 0$ once $|x|$ is small. Near its minimizer x^4 is too flat for a linear rate.

5.4 Condition number and iteration count

With $\alpha = 1/L$ the guarantee reads $f(x_k) - f^* \le (1 - 1/\kappa)^k\big(f(x_0) - f^*\big)$. To reduce the gap by a factor ε it suffices that $(1 - 1/\kappa)^k \le \varepsilon$, that is,

$$k \ge \frac{\ln \varepsilon}{\ln(1 - 1/\kappa)},$$

where both logarithms are negative. Since $\ln(1 - 1/\kappa) \le -1/\kappa$, the right-hand side is at most $\kappa \ln(1/\varepsilon)$, and for large κ it is close to it. The count grows linearly in κ and only logarithmically in the accuracy: this is the $O(\kappa \log(1/\varepsilon))$ of the lecture.

Example. Take $\kappa = 1000$ and $\varepsilon = 10^{-6}$. Then $\ln \varepsilon / \ln(1 - 1/\kappa)$ rounds up to $k = 13809$, and the estimate $\kappa \ln(1/\varepsilon)$ gives about 13816. For a quadratic the optimal step $2/(L+\mu)$ multiplies every eigencomponent of the error by a number of absolute value at most $\rho = (\kappa - 1)/(\kappa + 1)$, so the gap shrinks at least by ρ^2 per step, and the same reduction needs

$$k \ge \frac{\ln \varepsilon}{2 \ln \rho},$$

that is, $k = 3454$. This is about a quarter of the count at $1/L$, because $\ln \rho \approx -2/\kappa$ for large κ.

These counts are guarantees, and a run can beat them by far. In A2, with $\kappa = 2$, the guarantee gives 20 iterations and the run needs 10; in A3, with $\kappa = 50$, it gives 684 and the run needs 245. Two effects explain the difference. On a quadratic at $\alpha = 1/L$ the i-th eigencomponent of the gap shrinks by $(1 - \lambda_i/L)^2$ per step, so the slowest one shrinks by $(1 - \mu/L)^2$, the square of the guaranteed factor; this alone halves the count. And the stiff component is removed in the first step, since its factor is $1 - L/L = 0$: in A3, $f(x_0) = 25.5$ but $f(x_k) = \frac{1}{2}\,(0.98)^{2k}$ for $k \ge 1$.

With the optimal step $2/(L + \mu) = \frac{2}{51}$ both components of A3 shrink by exactly $\frac{49}{51}$ in absolute value, the bound $f(x_k) \le \rho^{2k} f(x_0)$ holds with equality, and bound and run agree: 173 iterations in both cases.

Check yourself. At fixed κ, what happens to the count when ε is replaced by ε^2?

Answer. It doubles: the count is proportional to $\ln(1/\varepsilon)$, and $\ln(1/\varepsilon^2) = 2\ln(1/\varepsilon)$. Twice as many orders of magnitude of the gap cost twice as many iterations; for large κ a condition number twice as large roughly doubles the count as well.

6 Reverse-mode automatic differentiation

Gradient descent needs ∇f at every step. **Automatic differentiation** (AD) computes it from the program that evaluates f, exactly up to rounding, by applying the chain rule to one elementary operation at a time. Its reverse mode delivers the whole gradient for a small constant multiple of the cost of one evaluation of f, however many variables there are. Everything here is done by hand on small graphs.

6.1 A computation as a graph

A program evaluates f through a sequence of intermediate variables, each an elementary function of earlier ones. For the example of the lecture, $z = \sin(xy) + x^2$:

$$u = xy, \qquad v = \sin u, \qquad w = x^2, \qquad z = v + w.$$

The nodes are the inputs x, y and the intermediates u, v, w, z; an edge $a \to b$ means that b uses a. Every variable uses only earlier ones, so there are no cycles: the graph is a **directed acyclic graph** (DAG). The order x, y, u, v, w, z in which the program computes the nodes is a **topological order**: every edge points forward in it.

Each node knows its local partial derivatives: $\partial u/\partial x = y$, $\partial u/\partial y = x$, $\partial v/\partial u = \cos u$, $\partial w/\partial x = 2x$, $\partial z/\partial v = \partial z/\partial w = 1$. AD combines these numbers by the chain rule; it never builds a formula for $\partial z/\partial x$ as a whole.

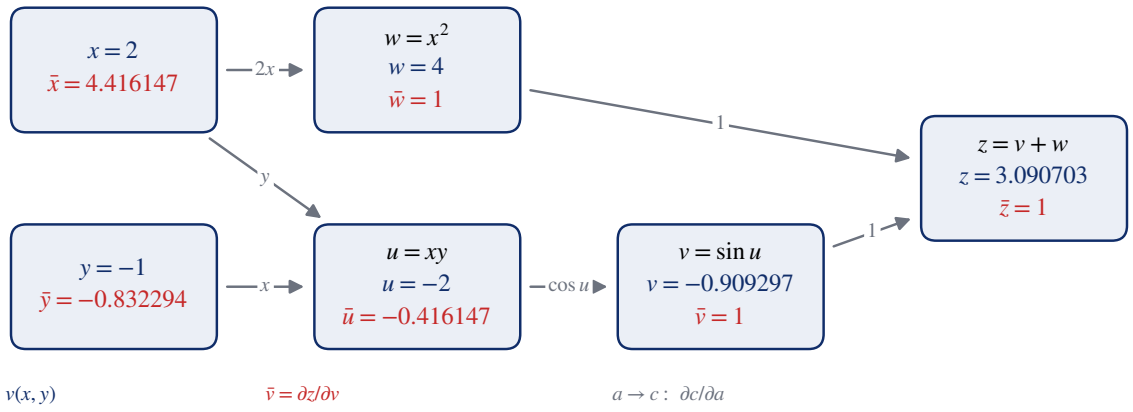


Figure. The graph of C1 at $(x, y) = (2, -1)$. Forward values: $u = -2$, $v = -0.909297$, $w = 4$, $z = 3.090703$. Adjoints of the reverse pass: $\bar{z} = \bar{v} = \bar{w} = 1$, $\bar{u} = \cos u = -0.416147$, $\bar{x} = 4.416147$, $\bar{y} = -0.832294$.

6.2 Forward mode

Choose a direction $(\dot{x}, \dot{y})$ of the inputs and carry, next to every value, its **tangent**: the derivative of that value along the chosen direction. The tangents follow the graph forward with the same local derivatives:

$$\dot{u} = y\dot{x} + x\dot{y}, \qquad \dot{v} = \cos(u)\,\dot{u}, \qquad \dot{w} = 2x\,\dot{x}, \qquad \dot{z} = \dot{v} + \dot{w}.$$

For a map $F : \mathbb{R}^n \to \mathbb{R}^m$ with Jacobian J, one forward sweep with the seed $\dot{x} \in \mathbb{R}^n$ returns the **Jacobian–vector product** $J\dot{x}$. For a scalar f this is the directional derivative $\nabla f(x)^\top \dot{x}$, a single number, so the full gradient needs n sweeps, one per basis vector $e_1, \dots, e_n$.

Example. At the point of C1 with the seed $\dot{x} = 1$, $\dot{y} = 0$ one sweep gives $\dot{z} = y \cos u + 2x = 4.416147$, the partial derivative in x. The derivative in y needs a second sweep with $\dot{x} = 0$, $\dot{y} = 1$.

6.3 Reverse mode

For a scalar output z, the **adjoint** of a node v is $\bar{v} = \partial z / \partial v$, the derivative of the output with respect to that node through everything downstream of it. Start with $\bar{z} = 1$ and every other adjoint equal to 0, visit the nodes in reverse topological order, and for every edge $u \to v$ add the contribution

$$\bar{u} \mathrel{+}= \bar{v} \frac{\partial v}{\partial u}.$$

When the sweep reaches a node, all nodes that use it have been processed, so its adjoint is complete. In one formula,

$$\bar{v}_i = \sum_{j:\ i \to j} \bar{v}_j \frac{\partial v_j}{\partial v_i},$$

the sum over all nodes j that use v_i.

In the example: $\bar{z} = 1$; the node $z = v + w$ gives $\bar{v} = 1$ and $\bar{w} = 1$; $w = x^2$ gives $\bar{x} \mathrel{+}= 2x$; $v = \sin u$ gives $\bar{u} = \cos u$; $u = xy$ gives $\bar{x} \mathrel{+}= y \cos u$ and $\bar{y} \mathrel{+}= x \cos u$. Hence

$$\bar{x} = y \cos u + 2x = \frac{\partial z}{\partial x}, \qquad \bar{y} = x \cos u = \frac{\partial z}{\partial y}.$$

One backward sweep has produced both partial derivatives.

For $F : \mathbb{R}^n \to \mathbb{R}^m$ a reverse sweep started from a seed $\bar{z} \in \mathbb{R}^m$ returns the **vector–Jacobian product** $\bar{z}^\top J$. For a scalar loss, $m = 1$ and $\bar{z} = 1$, this is the whole gradient in one sweep, whatever n is. Reverse mode applied to a model with a scalar loss and many parameters is **backpropagation**.

6.4 Why adjoints accumulate

A node that feeds several others receives a contribution along each outgoing edge, and the multivariable chain rule adds them. For $z = x^2 + x^3$ with the nodes $a = x^2$, $b = x^3$, $z = a + b$:

$$\frac{dz}{dx} = \frac{\partial z}{\partial a}\frac{da}{dx} + \frac{\partial z}{\partial b}\frac{db}{dx} = 2x + 3x^2.$$

This is why the rule is written with $+=$: the adjoint $\bar{x}$ collects $2x$ from a and $3x^2$ from b. An implementation that assigns instead of adding keeps only the contribution of the edge it visits last, $2x$ or $3x^2$, and the wrong answer even depends on the visit order.

In general $\partial z / \partial x$ is the sum, over all paths from x to z, of the products of the local derivatives along the path. The reverse sweep computes this sum without listing the paths: the adjoint of each node already holds the sum over all paths from that node to z. In C1 the input x feeds u and w, so $\bar{x}$ has two terms; in C2 it feeds both xy and $xy + x$.

Trap. Frameworks accumulate parameter gradients across calls for the same reason. Before a new gradient is computed, the stored adjoints must be reset to zero, or the new gradient is added to the old one.

6.5 Cost and memory

Let $f : \mathbb{R}^n \to \mathbb{R}$ be computed by N elementary operations. A forward or a reverse sweep does a bounded amount of work per operation, so each costs a small constant multiple of one evaluation of f. The difference between the modes is the number of sweeps.

Property	Forward mode	Reverse mode
Seed	a direction $\dot{x} \in \mathbb{R}^n$	an output weight $\bar{z} \in \mathbb{R}^m$
One sweep gives	$J\dot{x}$, a Jacobian–vector product	$\bar{z}^\top J$, a vector–Jacobian product
Gradient of $f : \mathbb{R}^n \to \mathbb{R}$	n sweeps	one forward and one reverse sweep
Jacobian of $F : \mathbb{R}^n \to \mathbb{R}^m$	n sweeps	m sweeps
Extra memory	one tangent per value in use	every intermediate value (the tape)

Reverse mode is cheap in operations but not in memory. The local derivatives need the forward values ($\partial v/\partial u = \cos u$ needs u), so the forward sweep must record every intermediate value, the **tape**, before the reverse sweep can start, and the memory grows with the length of the computation.

Checkpointing trades computation for memory. The forward sweep stores only some intermediate values, the checkpoints; during the reverse sweep each segment between two checkpoints is recomputed from the earlier one when its values are needed. With about $\sqrt{N}$ checkpoints spaced $\sqrt{N}$ operations apart, the memory drops to the order of $\sqrt{N}$ stored values at the price of roughly one extra forward sweep.

6.6 Backpropagation for the logistic loss

One observation has features $x \in \mathbb{R}^d$ and a label $y \in \{0, 1\}$. The model computes

$$s = w^\top x + b, \qquad p = \sigma(s) = \frac{1}{1 + e^{-s}}, \qquad \ell = -y \log p - (1 - y) \log(1 - p).$$

Since $\log p = s - \log(1 + e^s)$ and $\log(1 - p) = -\log(1 + e^s)$, the loss is $\ell = \log(1 + e^s) - ys$, the form used in C3.

The reverse pass starts with $\bar{\ell} = 1$. The node $\ell(s)$ gives

$$\bar{s} = \frac{d\ell}{ds} = \frac{e^s}{1 + e^s} - y = \sigma(s) - y,$$

and the node $s = w^\top x + b$ passes it on to the parameters: $\bar{w} = \bar{s}\, x$ and $\bar{b} = \bar{s}$. So

$$\nabla_w \ell = \big(\sigma(s) - y\big)\, x, \qquad \frac{\partial \ell}{\partial b} = \sigma(s) - y,$$

and the forward value $p = \sigma(s)$ is reused. For the mean loss $\frac{1}{n}\sum_i \ell_i$ the contributions of the observations are added and divided by n: $\nabla_w \frac{1}{n}\sum_i \ell_i = \frac{1}{n}\sum_i \big(\sigma(s_i) - y_i\big)\, x_i$.

With labels $\tilde{y} = 2y - 1 \in \{-1, +1\}$ the same loss reads

$$\ell = \log\big(1 + e^{-\tilde{y}s}\big).$$

The two cases check it: $y = 1$, $\tilde{y} = 1$ gives $\log(1 + e^{-s}) = \log(1 + e^s) - s$, and $y = 0$, $\tilde{y} = -1$ gives $\log(1 + e^s)$. Its derivative $d\ell/ds = -\tilde{y}\,\sigma(-\tilde{y}s)$ equals $\sigma(s) - y$ in both cases. The two conventions describe one function.

Trap. $\sigma(s) - y$ is the derivative with respect to s, not with respect to w. The gradient in w carries the factor x; without it the result is a number where a vector in $\mathbb{R}^d$ is expected.

6.7 Symbolic, numerical and automatic differentiation

Method	Core idea	Accuracy	Typical issue	Cost of a gradient in $\mathbb{R}^n$
Symbolic	manipulates the formula into a formula for the derivative	exact	expression swell: the derivative formula can be far larger than the program	evaluating that formula, whose size is not controlled
Finite differences	perturbs the inputs and divides differences of values	approximate, at best about $\varepsilon^{2/3}$ relative for central differences, with ε the machine epsilon	truncation for large steps, cancellation for small ones	$2n$ evaluations of f for central differences
Automatic, reverse mode	applies the chain rule to the program, node by node	exact up to rounding	memory for the tape; control flow and non-smooth operations	a small constant multiple of one evaluation of f

The central difference

$$\frac{\partial f}{\partial x_i}(x) \approx \frac{f(x + he_i) - f(x - he_i)}{2h}$$

has two errors. Taylor's formula $f(x \pm he_i) = f \pm h\,\partial_i f + \frac{h^2}{2}\partial_i^2 f \pm \frac{h^3}{6}\partial_i^3 f + O(h^4)$ makes the even terms cancel, so the **truncation** error is $\frac{h^2}{6}|\partial_i^3 f| + O(h^3)$, of order h^2. Each computed value of f carries a **round-off** error of about $\varepsilon|f|$, where ε is the machine epsilon of double precision. The difference of two nearly equal numbers keeps these absolute errors, and dividing by $2h$ gives an error of order $\varepsilon|f|/h$. The total

$$E(h) \approx \frac{h^2}{6}M + \frac{\varepsilon|f|}{h}, \qquad M = |\partial_i^3 f|,$$

is smallest where $E'(h) = \frac{hM}{3} - \frac{\varepsilon|f|}{h^2} = 0$, at $h^3 = 3\varepsilon|f|/M$. So the best step is of order $\varepsilon^{1/3} \approx 6{\cdot}10^{-6}$ and the best error of order $\varepsilon^{2/3}$. The one-sided difference $\big(f(x + he_i) - f(x)\big)/h$ has truncation error of order h, and the same argument gives the best step of order $\sqrt{\varepsilon}$.

Demo D3 measures this for $\partial z/\partial x$ of C1 at $(x, y) = (2,\ -1)$. The relative error falls like h^2 from $1.6 \cdot 10^{-4}$ at $h = 10^{-1}$, reaches its minimum $5.7 \cdot 10^{-12}$ at $h = 10^{-5}$, close to $\varepsilon^{1/3}$, and grows again to $1.4 \cdot 10^{-7}$ at $h = 10^{-10}$ as round-off takes over.

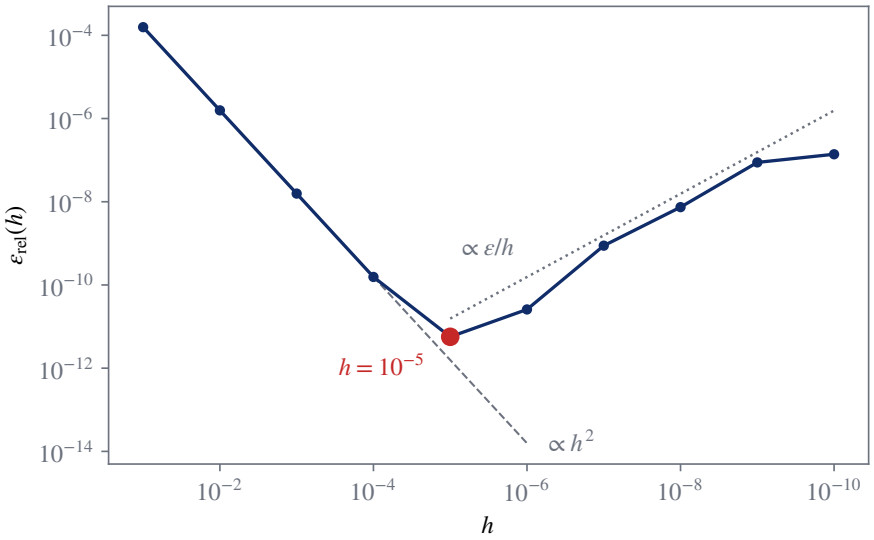


Figure. Relative error of the central difference for $\partial z/\partial x$ of C1 at $(x, y) = (2,\ -1)$, for h from 10^{-1} to 10^{-10}. For large h the truncation error dominates and falls as h^2; for small h round-off dominates and grows as $1/h$.

AD has no step to choose and no truncation error: it evaluates the exact derivative of the program in floating point. Finite differences remain useful as an independent check of a gradient, at a few points, with h near $\varepsilon^{1/3}$ and a relative error measure.

6.8 Problems of block C

The three seminar problems on the reverse pass, with complete solutions.

C1. The lecture's graph at a point

On the board

Statement. Take $z = \sin(xy) + x^2$ at $(x, y) = (2,\ -1)$ with the intermediate nodes $u = xy$, $w = x^2$, $v = \sin u$. List the nodes in topological order, run the forward pass, then the reverse pass from $\bar{z} = 1$. Report z, $\partial z/\partial x$ and $\partial z/\partial y$, and check both derivatives analytically.

Solution.

1. Topological order, every node after its inputs: x, y, $u = xy$, $w = x^2$, $v = \sin u$, $z = v + w$. The input x has two outgoing edges, into u and into w.
2. Forward pass at $(x, y) = (2,\ -1)$, keeping every value:

$$u = -2, \quad w = 4, \quad v = \sin(-2) = -0.909297, \quad z = v + w = 3.090703.$$

   The reverse pass needs u, x and y again.
3. Reverse pass, node $z = v + w$: seed $\bar{z} = 1$. Both local derivatives are 1, so $\bar{v} = 1$ and $\bar{w} = 1$.
4. Node $v = \sin u$: with u in radians, $\bar{u}\ \mathrel{+}= \bar{v}\cos u = \cos(-2) = -0.416147$.
5. Node $w = x^2$: the local derivative is $2x$, so $\bar{x}\ \mathrel{+}= \bar{w} \cdot 2x = 2 \cdot 2$. This is the first of the two contributions to $\bar{x}$.
6. Node $u = xy$: the local derivatives are y and x, so

$$\bar{x}\ \mathrel{+}= \bar{u}\, y = (-0.416147)(-1) = 0.416147, \qquad \bar{y}\ \mathrel{+}= \bar{u}\, x = (-0.416147) \cdot 2 = -0.832294.$$

   With both contributions $\bar{x} = 0.416147 + 2 \cdot 2 = 4.416147$; overwriting would leave only 0.416147.

7. Analytic check: $\partial z/\partial x = y\cos(xy) + 2x$ and $\partial z/\partial y = x\cos(xy)$ give the same numbers. The reverse pass evaluated these formulas term by term: $\bar{u}\,y$ is $y\cos(xy)$ and $\bar{w}\cdot 2x$ is $2x$. One sweep produced both partial derivatives.

Answer. $z = 3.090703$, $\partial z/\partial x = 4.416147$, $\partial z/\partial y = -0.832294$.

C2. An input used twice

On your own

Statement. Take $q = (xy + x)^2$ at $(x, y) = (2,\ 3)$ with the nodes $a = xy$, $b = a + x$, $q = b^2$. Compute q by a forward pass and $\partial q/\partial x$, $\partial q/\partial y$ by one reverse pass, with pen and paper.

Solution.

1. Topological order: x, y, $a = xy$, $b = a + x$, $q = b^2$. Two edges leave x: one into a and one directly into b.

2. Forward pass at $(x, y) = (2,\ 3)$: $a = xy = 6$, $b = a + x = 8$, $q = b^2 = 64$.

3. Reverse pass: $\bar{q} = 1$, and the node $q = b^2$ gives $\bar{b} = \bar{q} \cdot 2b = 16$.

4. Node $b = a + x$: both local derivatives are 1, so $\bar{a}$ += $\bar{b} = 16$ and $\bar{x}$ += $\bar{b} = 16$.

5. Node $a = xy$: $\bar{x}$ += $\bar{a}\,y = 48$ and $\bar{y}$ += $\bar{a}\,x = 32$. So $\bar{x} = 16 + 48 = 64$; with `=` in place of `+=` it would be 48, the last contribution only.

6. Analytic check: $q = x^2(y+1)^2$, so $\partial q/\partial x = 2x(y+1)^2 = 64$ and $\partial q/\partial y = 2x^2(y+1) = 32$. The value q and $\partial q/\partial x$ agree here only because $x = 2$: their ratio is $x/2$.

Answer. $q = 64$, $\partial q/\partial x = 48 + 16 = 64$, $\partial q/\partial y = 32$.

C3. The logistic loss node

On your own

Statement. For one example with score s and label $y \in \{0, 1\}$ the logistic loss is $\ell(s) = \log(1 + e^s) - ys$, and $\sigma(s) = 1/(1 + e^{-s})$. At $s = 0.5$ and $y = 1$, split ℓ into elementary nodes, run the forward and the reverse pass, and report ℓ, $\sigma(s)$ and $d\ell/ds$. Then rewrite the loss for labels $\tilde{y} = 2y - 1 \in \{-1, 1\}$ as a function of $\tilde{y}s$ alone, and check it for both labels.

Solution.

1. Nodes in topological order: s, $u = e^s$, $v = 1 + u$, $w = \log v$, $\ell = w - ys$. The label y is a constant. The score s feeds two nodes, u and ℓ, so $\bar{s}$ will have two terms.

2. Forward pass at $s = 0.5$, $y = 1$:

$$u = e^s = 1.648721, \quad v = 1 + 1.648721, \quad w = \log v = 0.974077,$$

$$\ell = w - ys = 0.974077 - 1 \cdot 0.5 = 0.474077.$$

3. Reverse pass: $\bar{\ell} = 1$. The node $\ell = w - ys$ has local derivative 1 in w and $-y$ in s, so $\bar{w} = 1$ and $\bar{s}$ += $-y = -1$.

4. Through the logarithm: $\bar{v} = \bar{w}/v = \dfrac{1}{1 + e^s}$, then $\bar{u} = \bar{v}$, then

$$\bar{s} \mathrel{+}= \bar{u}\,e^s = \frac{e^s}{1 + e^s} = \frac{1}{1 + e^{-s}} = \sigma(s) = 0.622459.$$

5. Both terms together: $d\ell/ds = \bar{s} = \sigma(s) - y = 0.622459 - 1 = -0.377541$. This is the lecture's $p - y$ with $p = \sigma(s)$. It is negative, so a gradient step raises the score of a positive example.

6. Labels ± 1: put $\tilde{y} = 2y - 1$. For $y = 1$, $\tilde{y} = 1$ and $\log(1 + e^s) - s = \log \dfrac{1 + e^s}{e^s} = \log(1 + e^{-s})$. For $y = 0$, $\tilde{y} = -1$ and the loss is $\log(1 + e^s)$ unchanged. Both cases read

$$\ell = \log\left(1 + e^{-\tilde{y}s}\right).$$

7. Check at the point: $\tilde{y} = 1$ and $\log(1 + e^{-\tilde{y}s}) = 0.474077$, the same number as ℓ. The two conventions are one function, so they share every value and the derivative $\sigma(s) - y$. The lecture writes the loss with labels in $\{0, 1\}$, and the ± 1 form is the other common convention.

Answer. $\ell = 0.474077$, $\sigma(s) = 0.622459$, $d\ell/ds = \sigma(s) - y = -0.377541$; with $\tilde{y} = 2y - 1$ the loss is $\log(1 + e^{-\tilde{y}s})$.

7 Colloquium

At the colloquium definitions come first, so the checklist of definitions from the first section comes first here too. Then come the four proofs with their key inequalities and three ticket-style problems with full solutions.

7.1 Definitions checklist

For each definition give the name, the formula and the quantifiers: for which points, or pairs of points, the formula must hold.

- Local minimizer: x^* such that there is $\varepsilon > 0$ with $f(x) \geq f(x^*)$ for all x with $\|x - x^*\| < \varepsilon$.
- Global minimizer: x^* such that $f(x) \geq f(x^*)$ for all x.
- Stationary point: a point x at which f is differentiable and $\nabla f(x) = 0$.
- Convex set: $(1 - t)x + ty \in C$ for all $x, y \in C$ and $t \in [0, 1]$.
- Convex function (Jensen): $f\big((1 - t)x + ty\big) \leq (1 - t)f(x) + tf(y)$ for all x, y and $t \in [0, 1]$.
- Strictly convex function: the same inequality is strict for all $x \neq y$ and $t \in (0, 1)$.
- μ-strongly convex function, $\mu > 0$: $f(y) \geq f(x) + \nabla f(x)^\top (y - x) + \frac{\mu}{2}\|y - x\|^2$ for all x, y; equivalently, $f - \frac{\mu}{2}\|\cdot\|^2$ is convex.
- L-smooth function: $\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|$ for all x, y.
- Condition number: $\kappa = L/\mu \geq 1$; for $f = \frac{1}{2}x^\top Q x$ it is $\lambda_{\max}(Q)/\lambda_{\min}(Q)$.
- Descent direction at x: d with $\nabla f(x)^\top d < 0$.
- PL condition with $\mu > 0$: $\frac{1}{2}\|\nabla f(x)\|^2 \geq \mu\big(f(x) - f^*\big)$ for all x.

7.2 Four proofs

Descent lemma
If ∇f is L-Lipschitz, then $f(y) \leq f(x) + \nabla f(x)^\top (y - x) + \frac{L}{2}\|y - x\|^2$. Put $r(t) = f(x + t(y - x))$, write $f(y) - f(x) = \int_0^1 r'(t)\, dt$ with $r'(t) = \nabla f(x + t(y - x))^\top (y - x)$, subtract $\nabla f(x)^\top (y - x)$

and bound the rest by Cauchy–Schwarz and the Lipschitz condition:

$$f(y) - f(x) - \nabla f(x)^\top (y - x) = \int_0^1 \big(\nabla f(x + t(y - x)) - \nabla f(x)\big)^\top (y - x)\, dt \le \int_0^1 Lt\|y - x\|^2\, dt = \frac{L}{2}\|y - x\|^2.$$

Armijo termination

If ∇f is L-Lipschitz and $g = \nabla f(x) \neq 0$, backtracking along $d = -g$ stops after finitely many reductions. The descent lemma at $y = x - \alpha g$ gives

$$f(x - \alpha g) \le f(x) - \alpha\Big(1 - \frac{L\alpha}{2}\Big)\|g\|^2,$$

and the right-hand side is at most $f(x) - c\alpha\|g\|^2$ as soon as $1 - L\alpha/2 \ge c$, that is, for every $\alpha \le 2(1 - c)/L$. The trials $\beta^j \alpha_0$ tend to 0, so one of them enters this interval. The accepted step is either α_0 or β times a rejected trial, and a rejected trial is larger than $2(1 - c)/L$, so $\alpha \ge \min\{\alpha_0,\ 2\beta(1 - c)/L\}$.

Rate for convex functions

For convex L-smooth f and $\alpha = 1/L$, $f(x_k) - f^* \le L\|x_0 - x^*\|^2/(2k)$. Combine the descent step $f(x_{k+1}) \le f(x_k) - \frac{1}{2L}\|g_k\|^2$, convexity $f(x_k) - f^* \le g_k^\top (x_k - x^*)$ and the three-point identity for $x_{k+1} = x_k - \frac{1}{L} g_k$ into the key inequality

$$f(x_{k+1}) - f^* \le \frac{L}{2}\Big(\|x_k - x^*\|^2 - \|x_{k+1} - x^*\|^2\Big).$$

Sum it over the first k steps, where the right-hand side telescopes, and use that $f(x_j)$ does not increase.

Linear rate under strong convexity and under PL

Strong convexity implies PL: minimize the lower bound $f(x) + \nabla f(x)^\top (y - x) + \frac{\mu}{2}\|y - x\|^2$ over y to get $f^* \ge f(x) - \frac{1}{2\mu}\|\nabla f(x)\|^2$. Under PL and $\alpha = 1/L$ the key inequality is

$$f(x_{k+1}) - f^* \le f(x_k) - f^* - \frac{1}{2L}\|\nabla f(x_k)\|^2 \le \Big(1 - \frac{\mu}{L}\Big)\big(f(x_k) - f^*\big),$$

and iterating it gives $(1 - \mu/L)^k$. The second proof uses only PL, not convexity.

7.3 Ticket T1 on conditioning and rates

Check yourself. Let $f(x) = \frac{1}{2} x^\top Q x$ with $Q = \mathrm{diag}(4,\ 100)$. Find μ, L and κ. For which constant steps does gradient descent converge from every starting point? How many iterations does the guarantee at $\alpha = 1/L$ need to reduce $f(x_k) - f^*$ by the factor 10^{-6}? Find the optimal constant step, its contraction factor and the number of iterations it needs for the same reduction.

Answer. $\mu = \lambda_{\min}(Q) = 4$, $L = \lambda_{\max}(Q) = 100$, $\kappa = L/\mu = 25$. In eigen-coordinates the iteration is $x_{k+1,i} = (1 - \alpha\lambda_i) x_{k,i}$, so it converges from every start exactly when $|1 - \alpha\lambda_i| < 1$ for both i, that is, for $0 < \alpha < 2/L = 0.02$. At $\alpha = 1/L = 0.01$ the guarantee $f(x_k) - f^* \le (1 - \mu/L)^k \big(f(x_0) - f^*\big)$ has the factor $1 - \mu/L = 0.96$, and $(0.96)^k \le 10^{-6}$ requires $k \ge \ln(10^{-6})/\ln(0.96)$, so $k = 339$. The optimal constant step minimizes the worst factor

$\max\{|1-\alpha\mu|,\ |1-\alpha L|\}$. The first term decreases in α and the second increases once $\alpha > 1/L$, so the minimum is where $1-\alpha\mu = \alpha L - 1$:

$$\alpha^* = \frac{2}{L+\mu} = \tfrac{1}{52} \approx 0.019231, \qquad \rho = \frac{L-\mu}{L+\mu} = \frac{\kappa-1}{\kappa+1} = \tfrac{12}{13} \approx 0.923077.$$

Both components of x_k shrink by exactly ρ in absolute value, so $f(x_k) = \rho^{2k} f(x_0)$, and $\rho^{2k} \le 10^{-6}$ requires $k \ge \ln(10^{-6})/(2\ln\rho)$, so $k = 87$: almost four times fewer iterations than the guarantee at $1/L$.

7.4 Ticket T2 on Armijo backtracking

Check yourself. Run Armijo backtracking on $f(x) = \frac{1}{2}x^\top \operatorname{diag}(2,\ 20)\,x$ from $x_0 = (1,\ 1)$ along $d = -\nabla f(x_0)$ with $\alpha_0 = 1$, $\beta = 0.5$ and $c = 0.25$. List the trials, find the accepted step and the number of halvings, and check the result against the termination theorem.

Answer. Here $L = \lambda_{\max} = 20$, $f(x_0) = 11$, $g = \nabla f(x_0) = (2,\ 20)$ and $\|g\|^2 = 404$. A trial α is accepted when $f(x_0 - \alpha g) \le f(x_0) - c\alpha\|g\|^2$:

α	$f(x_0-\alpha g)$	$f(x_0) - c\alpha\|g\|^2$	accepted
1	3611	-90	no
0.5	810	-39.5	no
0.25	160.25	-14.25	no
0.125	23.0625	-1.625	no
0.0625	1.390625	4.6875	yes

The loop stops after 4 halvings with $\alpha = 0.0625$. The theorem guarantees acceptance for every $\alpha \le 2(1-c)/L = 0.075$, so the loop cannot go past the first trial below this value, which is 0.0625. Since at least one trial was rejected, the accepted step also satisfies $\alpha \ge \beta \cdot 2(1-c)/L = 0.0375$. Both checks agree with the run: $0.0375 \le 0.0625 \le 0.075$.

7.5 Ticket T3 on a reverse pass

Check yourself. Compute $f(x,y) = xe^y + y^2$ and both partial derivatives at $(x,y) = (2,\ 1)$ by a reverse pass: list the nodes in topological order, compute the forward values, then the adjoints, and check the result against the analytic gradient.

Answer. Nodes in topological order: the inputs x, y, then $a = e^y$, $b = xa$, $c = y^2$, $f = b + c$. Forward values: $a = e \approx 2.718282$, $b = 5.436564$, $c = 1$, $f = 6.436564$. Reverse pass from $\bar f = 1$: the node $f = b + c$ gives $\bar b = 1$ and $\bar c = 1$; $c = y^2$ gives $\bar y \mathrel{+}= \bar c \cdot 2y = 2$; $b = xa$ gives $\bar x \mathrel{+}= \bar b\, a = 2.718282$ and $\bar a \mathrel{+}= \bar b\, x = 2$; $a = e^y$ gives $\bar y \mathrel{+}= \bar a\, e^y = 5.436564$. The input y feeds two nodes, so its adjoint accumulates: $\bar y = 2 + 5.436564 = 7.436564$. Analytically $\partial f/\partial x = e^y = e$ and $\partial f/\partial y = xe^y + 2y = 2e + 2$, which at this point are 2.718282 and 7.436564. Overwriting $\bar y$ instead of adding would give 2 or 5.436564, depending on the order.

8 Useful constants

The step $1/L$, the stability boundary $2/L$ and the rate $1 - \mu/L$ all need the constants of the problem. For a quadratic they are eigenvalues; for logistic regression L has a simple upper bound.

8.1 Logistic regression

The data are $x_1, \dots, x_n \in \mathbb{R}^d$, the rows of $X \in \mathbb{R}^{n \times d}$, with labels $\tilde{y}_i \in \{-1, +1\}$. The mean logistic loss is

$$f(w) = \frac{1}{n} \sum_{i=1}^{n} \log\big(1 + e^{-\tilde{y}_i w^\top x_i}\big),$$

or, in the $\{0, 1\}$ form, $\frac{1}{n}\sum_i \big(\log(1 + e^{s_i}) - y_i s_i\big)$ with $s_i = w^\top x_i$ and $\tilde{y}_i = 2y_i - 1$; the two are the same function. A bias is absorbed into w by a constant feature 1 in every x_i.

Proposition: Smoothness of the logistic loss. With $\sigma_i = \sigma(w^\top x_i)$,

$$\nabla^2 f(w) = \frac{1}{n} X^\top D X, \qquad D = \operatorname{diag}\big(\sigma_i(1 - \sigma_i)\big),$$

and $0 \preceq \nabla^2 f(w) \preceq \frac{1}{4n} X^\top X$ for every w. Hence f is convex and L-smooth with

$$L \le \frac{\lambda_{\max}(X^\top X)}{4n}.$$

Proof. In the $\{0, 1\}$ form the i-th term is $\varphi(s_i) - y_i s_i$ with $\varphi(s) = \log(1 + e^s)$, $\varphi'(s) = \sigma(s)$ and $\varphi''(s) = \sigma(s)\big(1 - \sigma(s)\big)$; the linear term has no curvature. By the chain rule with $s_i = w^\top x_i$ the Hessian of the i-th term is $\sigma_i(1 - \sigma_i)\, x_i x_i^\top$, and the mean of these matrices is $\frac{1}{n} X^\top D X$. Since $p(1 - p) \le \frac{1}{4}$ for $p \in [0, 1]$, with equality at $p = \frac{1}{2}$, for every v

$$0 \le v^\top \nabla^2 f(w)\, v = \frac{1}{n} \sum_i \sigma_i(1 - \sigma_i)(x_i^\top v)^2 \le \frac{1}{4n} \sum_i (x_i^\top v)^2 = \frac{1}{4n} \|Xv\|^2 \le \frac{\lambda_{\max}(X^\top X)}{4n} \|v\|^2.$$

A twice differentiable function with $0 \preceq \nabla^2 f \preceq LI$ everywhere is convex, and its gradient is L-Lipschitz because $\nabla f(y) - \nabla f(x) = \int_0^1 \nabla^2 f\big(x + t(y - x)\big)(y - x)\, dt$.

Here $\lambda_{\max}(X^\top X) = \|X\|_2^2$, the square of the largest singular value of X. The bound is attained at $w = 0$, where every $\sigma_i = \frac{1}{2}$ and $\nabla^2 f(0) = \frac{1}{4n} X^\top X$, so no smaller constant is valid on the whole space. The loss is not strongly convex: along $w = tv$ with $x_i^\top v \neq 0$ for all i the weights $\sigma_i(1 - \sigma_i)$ tend to 0 as $t \to \infty$, and so does the Hessian.

With a penalty, $f_\lambda(w) = f(w) + \lambda \|w\|^2$ with a weight $\lambda > 0$ (a number, not an eigenvalue), the Hessian gains $\nabla^2\big(\lambda \|w\|^2\big) = 2\lambda I$:

$$2\lambda I \preceq \nabla^2 f_\lambda(w) \preceq \Big(\frac{\lambda_{\max}(X^\top X)}{4n} + 2\lambda\Big) I.$$

So f_λ is 2λ-strongly convex and L-smooth with $L \le \frac{\lambda_{\max}(X^\top X)}{4n} + 2\lambda$, and

$$\kappa = \frac{L}{\mu} \le \frac{\lambda_{\max}(X^\top X)/(4n) + 2\lambda}{2\lambda} = 1 + \frac{\lambda_{\max}(X^\top X)}{8n\lambda}.$$

Gradient descent with $\alpha = 1/L$ then converges linearly, and a smaller λ means a larger bound on κ and a slower guaranteed rate.

Trap. Some texts write the penalty as $\frac{\lambda}{2}\|w\|^2$. Then the Hessian gains λI, not $2\lambda I$, the problem is λ-strongly convex, and every constant above changes accordingly.

8.2 The quadratics of the seminar

For $f(x) = \frac{1}{2}x^\top Qx$ with diagonal Q: $\mu = \lambda_{\min}(Q)$, $L = \lambda_{\max}(Q)$, gradient descent converges for $0 < \alpha < 2/L$, the linear guarantee holds at $\alpha = 1/L$, and the optimal constant step is $2/(L+\mu)$.

Problem	Q	μ	L	κ	$1/L$	$2/L$	$2/(L+\mu)$
A1	diag(1, 10)	1	10	10	0.1	0.2	$\frac{2}{11}$
A2	diag(1, 2)	1	2	2	0.5	1	$\frac{2}{3}$
A3	diag(1, 50)	1	50	50	0.02	0.04	$\frac{2}{51}$

B1 uses the matrix of A1; B2 and B3 use the matrix of A3.